



Role of Ceramides and Sphingolipids in Parkinson's Disease

Melissa Vos^{1*}, Christine Klein¹ and Andrew A Hicks²

¹ - Institute of Neurogenetics, University of Luebeck, 23562 Luebeck, Germany

² - Institute for Biomedicine (affiliated to the University of Luebeck, Luebeck, Germany), Eurac Research, 39100 Bolzano, Italy

Correspondence to Melissa Vos: *Institute of Neurogenetics, University of Luebeck, UKSH, Ratzeburger Allee 160, Building 67 (BMF), 23562 Luebeck, Germany. melissa.vos@neuro.uni-luebeck.de (M. Vos) @andrewhicks (A.A Hicks)

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Abstract

Sphingolipids, including the basic ceramide, are a subset of bioactive lipids that consist of many different species. Sphingolipids are indispensable for proper neuronal function, and an increasing number of studies have emerged on the complexity and importance of these lipids in (almost) all biological processes. These include regulation of mitochondrial function, autophagy, and endosomal trafficking, which are affected in Parkinson's disease (PD). PD is the second most common neurodegenerative disorder and is characterized by the loss of dopaminergic neurons. Currently, PD cannot be cured due to the lack of knowledge of the exact pathogenesis. Nonetheless, important advances have identified molecular changes in mitochondrial function, autophagy, and endosomal function. Furthermore, recent studies have identified ceramide alterations in patients suffering from PD, and in PD models, suggesting a critical interaction between sphingolipids and related cellular processes in PD. For instance, autosomal recessive forms of PD cause mitochondrial dysfunction, including energy production or mitochondrial clearance, that is directly influenced by manipulating sphingolipids. Additionally, endo-lysosomal recycling is affected by genes that cause autosomal dominant forms of the disease, such as *VPS35* and *SNCA*. Furthermore, endo-lysosomal recycling is crucial for transporting sphingolipids to different cellular compartments where they will execute their functions. This review will discuss mitochondrial dysfunction, defects in autophagy, and abnormal endosomal activity in PD and the role sphingolipids play in these vital molecular processes.

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Introduction

Parkinson's disease (PD) is a complex neurodegenerative disorder affecting 1–2/1000 of the population worldwide.¹ As the population ages, the number of patients suffering from PD is reportedly increasing exponentially,² exhibiting epidemic features that might go beyond the increase in patients with Alzheimer's disease, the most common neurodegenerative disorder. From 1990 to 2015, the number of people with PD doubled to 6 million, and is predicted to double again to 12 million by 2040.^{3–4}

Most PD cases are classified as idiopathic (90%). However, over the past decades, genes were identified that cause PD when mutated. These genetic forms of PD can be subdivided into autosomal dominant forms caused by mutations in *SNCA*, *LRRK2*, and *VPS35* and autosomal recessive forms caused by loss of *PINK1*, *Parkin*, or *DJ-1*.^{5–6} Studies using genetic animal models have revealed several molecular processes to be affected that converge into Parkinsonian-like phenotypes, that also provide insights into pathogenic mechanisms in idiopathic forms of PD. Pathologically, PD is hallmarked by

the presence of Lewy bodies and Lewy neurites, that are aggregations mostly existing of alpha-synuclein, along with lipid membrane and organelle fragments.⁷ Furthermore, PD is characterized by the loss of dopaminergic neurons resulting in bradykinesia, tremor, and rigidity.⁸ In addition, non-motor symptoms such as dementia, sleep deprivation, and depression appear.^{9–12} Currently, the disease cannot be halted or reversed due to the incomplete knowledge of the pathogenesis of PD, and so therapeutic strategies consist mainly of symptom treatment.¹³ Despite this, studies using genetic models of PD over recent years have dramatically increased our knowledge of the mechanisms underlying PD. Interestingly, genetic and sporadic forms of PD converge to similar signs and symptoms, enabling findings from genetic PD models to be relevant for idiopathic forms of PD. The outcome of these studies revealed mitochondrial dysfunction mainly linked to the autosomal recessive proteins, autophagy defects, and abnormalities in the endosomal system, more often linked to the autosomal dominant forms of PD.¹⁴

More recently, lipids have emerged to play a crucial role in the pathogenesis of PD. Interestingly, lipids have been found in Lewy bodies.⁷ Furthermore, variants in the sterol regulatory element-binding protein 1 (SREBP-1) act as a risk factor in fly models for PD via its regulation of mitophagy.¹⁵ In addition, cardiolipin, a mitochondrial-specific phospholipid, is decreased in Pink1-deficient *Drosophila melanogaster*. Elevating cardiolipin levels improves mitochondrial function in these *pink1*-mutant flies.¹⁶ Moreover, Stomatin-like protein-2 (SLP-2), that binds to cardiolipin and interacts with Parkin, and its increased expression, compensates for the mitochondrial dysfunction caused by loss of Parkin in *Drosophila melanogaster* and iPSC-derived neurons,¹⁷ suggesting cardiolipin (and its binding partners) can exert a protective role in PD. Finally, GBA, an enzyme involved in the synthesis of ceramide, the basic sphingolipid, is the most common risk factor for PD. Remarkably, ceramide levels are altered in PD patients^{18–19} and different genetic models of PD,^{20–22} suggesting a critical role for this lipid in the pathogenesis of PD.

This review will focus on the importance of sphingolipids (SL) in PD and its affected molecular processes. First, we will give an overview of sphingolipids and their metabolism. Next, we will review mitochondrial dysfunction, autophagy defects and endosome abnormalities in PD, focus on sphingolipids and discuss how these lipids function in the different PD-related mechanisms thereby contributing to the pathogenesis of Parkinson's disease.

Sphingolipids

Structure and synthesis

Sphingolipids are bioactive lipid species composed of a sphingosine backbone esterified to

a fatty acyl chain via an amino bond.²³ An enormous variety of sphingolipids exist depending on the head group that binds to the sphingoid base and can include phosphates, phosphocholine, saccharides, and variations in the fatty acyl chain depending on the length of the carbon chain and saturation and hydroxylation status.²⁴ Ceramide, the basic sphingolipid that forms the precursor for generating more complex sphingolipids, can be produced via several pathways, including *de novo* synthesis or as a breakdown or recycling product.

De novo synthesis of ceramide occurs in the endoplasmic reticulum (ER). It begins with the condensation of serine to palmitoyl CoA via serine palmitoyl transferase (SPT) activity to form 3-ketosphingosine, followed by the conversion to sphinganine and dihydroceramide, via ceramide synthase (CerS) activity, which is finally desaturated into ceramide.^{25–26} The first SPT step is rate-limiting in ceramide synthesis, and recent findings also show that in addition to palmitate, myristate or stearate can be used as a fatty acid substrate for SPT.^{27–28} Ceramide can then be further transformed into sphingomyelin via sphingomyelin synthase, or into sphingosine and sphingosine-1 phosphate via ceramidase and sphingosine kinase, and finally into glucosylceramide via glucosylceramide synthase and galactosylceramide via the galactosylceramide transferase^{29–30} (Figure 1).

Sphingolipid metabolism

Sphingolipid metabolism is a tightly regulated process that includes activity of approximately 40 different enzymes in mammals, and is determined by enzyme expression, post-translational modification, and allosteric mechanisms. In addition, it is dependent on the subcellular localization of key enzymes responsible for key synthetic steps. The sphingomyelin pathway, to generate sphingomyelin, occurs in the Golgi apparatus.^{26,31} Ceramide therefore needs to be transported from the ER to the Golgi in a non-vesicular manner, which is done via the ceramide transporter protein (CERT).^{32–33} Sphingosine is generated in the salvage pathway that occurs in the lysosomes where ceramide is reformed as the breakdown product of sphingomyelin via acid sphingomyelinase. Sphingosine can be further converted into sphingosine-1-phosphate via sphingosine kinases.^{26,29} In mitochondria, ceramide is generated via neutral sphingomyelinase or mitochondrially-localized CerS.^{34–36}

Regulation of sphingolipid function

The many different sphingolipid species have a functional role in most, if not all, cellular processes. Sphingolipid function is dependent on the chain length of the fatty acyl group. The length of this carbon chain is CerS-specific^{35,37}: CerS1

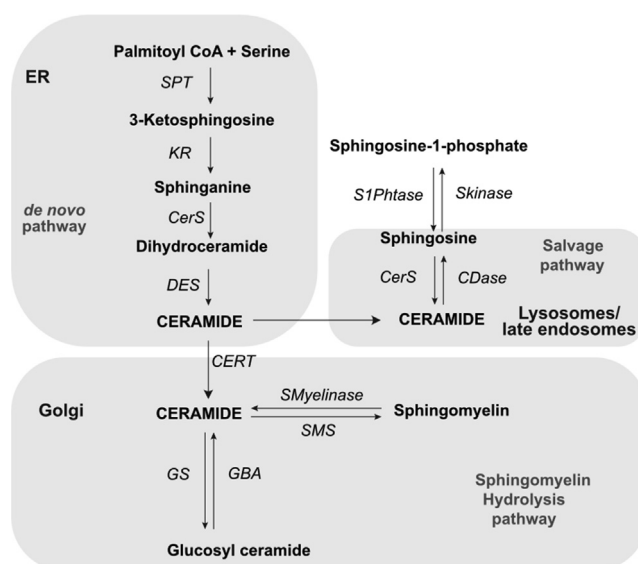


Figure 1. Scheme of the different biosynthetic pathways of ceramide that are located in the ER, lysosome, and Golgi complex. SPT: serine palmitoyl transferase; KR: 3-ketosphingosine reductase; CerS: ceramide synthase; DES: desaturase; SMylase: sphingomyelinase; SMS: sphingomyelin synthase; GBA: glucocerebrosidase; GS: glucosylceramide synthase; CDase: ceramidase; SKinase: sphingosine kinase; S1Phtase: sphingosine-1-phosphate phosphatase; CERT: ceramide transporter protein.

activity will mainly result in ceramide with a fatty acyl chain consisting of 18 carbons (C18-ceramide); CerS2 and CerS3 generate very long chain C22-24- and C26-ceramides, respectively; CerS4 provides C18- and C20-ceramides; and CerS5 and CerS6 generates primarily C16-ceramide, but can also generate C12- and C14-ceramides. All these different ceramide species can exert different and sometimes antagonistic functions,^{38–40} as was observed in oncogenic studies in which C18-ceramide provokes cell death in head and neck cancer, while C16-ceramide promotes cell proliferation in the same.⁴¹ This is also observed in Type II diabetes (T2D), where high plasma concentrations of Cer18:0 and Cer22:0 were associated with a higher T2D risk, while Cer20:0 was associated with lower T2D risk.⁴² Interestingly, cellular localization can also direct the function of ceramide. Mitochondrial accumulation of C18-ceramide induces ceramide-induced mitophagy, while C16-ceramide cannot; however, when C16-ceramide is targeted to mitochondria, it can provoke ceramide-induced mitophagy, suggesting the localization of ceramide production is a decisive factor for induction of mitophagy.⁴³

Challenges in the study of sphingolipids

Important advances have been made to study and identify the different sphingolipid species in different animal models, and the enzymes involved in sphingolipid metabolism are well known. Nonetheless, studying these sphingolipids is challenging. The variety of sphingolipid species and the specificity in cellular localization that

determine the function of sphingolipids, coupled with post-translational modifications that affect enzymatic activity, collectively complicate our full understanding of these lipids and their function, and demand caution when comparing findings that, at first glance, appear contradictory.

Mitochondrial Function and Sphingolipid Interaction in PD

Mitochondrial function

Mitochondria are the energy-generating organelles of the cell, and cells rich in mitochondria, such as neurons, usually have a high energy demand and need for calcium buffering. Energy is generated via the electron transport chain (ETC) (Figure 2). In this process electrons are provided from substrates NADH and FADH₂ from the Krebs cycle to complex I and II respectively, and are transported via ubiquinone to complex III. Here, cytochrome c will accept the electrons to be transferred to complex IV. During this transfer of electrons, protons are pumped from the matrix to the intermembrane space (IMS), creating a membrane potential across the inner mitochondrial membrane (IMM). This membrane potential will drive the ATPase (or complex V), in which protons are pumped inside the matrix to enable the conversion of ADP using O₂ into ATP and H₂O.^{44–45} One way to generate substrates for the Krebs cycle or the ETC is fatty acid oxidation or β -oxidation that occurs in the mitochondria. Fatty acids are imported to the mitochondria as fatty acyl

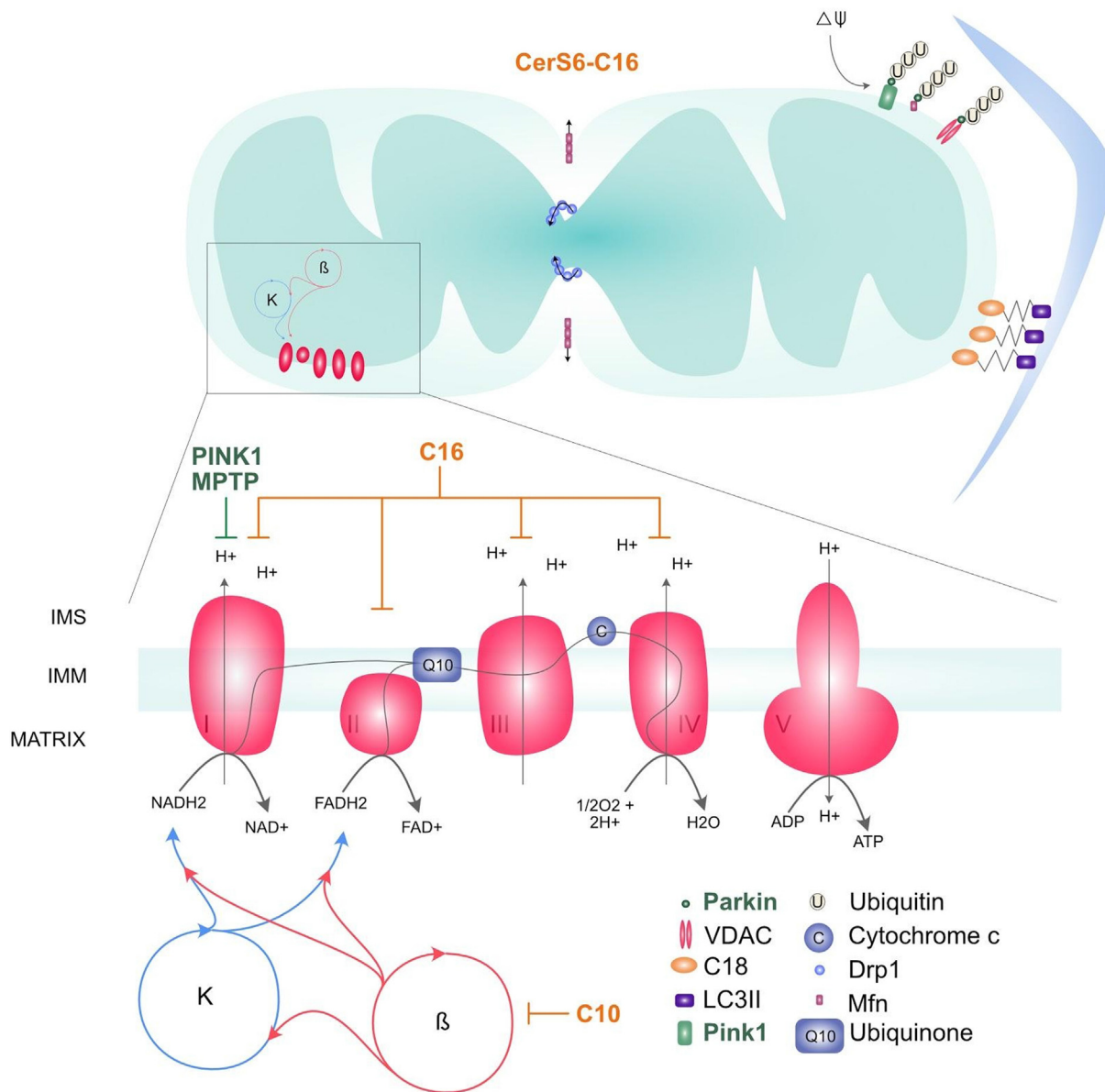


Figure 2. A simplified scheme of the implications of PD and sphingolipids on mitochondrial function. The ETC receives electrons from substrates via the Krebs cycle (K) and β -oxidation (β) that will be transported across the different complexes with the help of ubiquinone and cytochrome c. Loss of PINK1 or supplementation of MPTP inhibits complex I activity resulting in lower ATP levels. The sphingolipid C16 inhibits complex I-IV and C10 β -oxidation. Mitochondria undergo fusion (via Mfn) or fission (via Drp1). CerS-C16 binds to Mfn, and higher activity of Drp1 facilitates mitophagy that can be induced by a defective membrane potential through the PINK1-Parkin dependent mitophagy or via C18-induced mitophagy in which C18 directly binds to LC3 from the autophagosome. PD-related genes are highlighted in green. Drp1: dynamin-related protein 1; Mfn: mitofusin; IMM: inner mitochondrial membrane; IMS: intermembrane space.

CoA via a carnitine-dependent transport system. Here, two carbon atoms are removed from the fatty acyl chain to form acetyl CoA, which will enter the Krebs cycle, and NADH and FADH2 that are substrates for the ETC.⁴⁶

Mitochondria are dynamic organelles that undergo fusion and fission for which specific evolutionary conserved GTPase proteins are involved. For example, mitofusins (Mfn) induce

fusion of the outer mitochondrial membrane (OMM), and optic atrophy 1 (opa1) is responsible for the fusion of the IMM.^{47–48} Fission occurs by the Dynamin-related protein 1 (Drp1).^{48–49} Increased fission to fragment mitochondria facilitates mitophagy and therefore is beneficial to clear defective mitochondria.⁵⁰

Furthermore, mitochondria are important in calcium homeostasis as they are involved in

sequestering calcium in the IMM and releasing it when needed.⁵¹ In addition, mitochondria also provide ATP to the calcium ATPase in the plasma membrane.⁵² Calcium is important for neuronal function, as it is involved in vital processes such as neurotransmitter release and vesicle recycling.⁵¹ In mitochondria, calcium increases ATP production by changing enzymatic activity of calcium-sensitive enzymes in the mitochondrial matrix.^{53–55} However, excess mitochondrial calcium can lead to increased ROS generation, mTOR activity, and the accumulation of misfolded proteins, including α -synuclein aggregates, and thus enzymes such as the PD-GWAS-associated Inositol-1,4,5-triphosphate (IP3) kinase B (ITPKB) may play an important role in mitigating the negative consequences of excess mitochondrial calcium.⁵⁶

Mitochondrial defects in PD

The first indication of the involvement of mitochondria in the pathogenesis of PD came from studies of heroin abusers that developed irreversible Parkinsonism due to contamination with MPTP.^{57–58} Subsequent studies showed that MPTP specifically affects complex I of the ETC.^{59–61} Furthermore, decreased complex I function was observed in patients with PD.⁶² The identification of PINK1 as causal for PD when mutated provided the first direct link to mitochondrial dysfunction in PD. PINK1 is a serine/threonine kinase targeted to the mitochondria, where it is rapidly cleaved by PARL and imported via the TOM complex into the mitochondria.^{63–64} PINK1-dependent phosphorylation of NdufA10, a subunit of complex I, is required for proper complex I function.⁶⁵ Remarkably, stimulation of the ETC in *pink1*-mutant flies via vitamin K2⁶⁶ or near-infrared light⁶⁷ improves the *Pink1*-deficient fly phenotypes. The latter was later confirmed in a rat α -synuclein-dependent model.⁶⁸ Interestingly, loss of PINK1 impairs β -oxidation that can be improved following stimulation with carnitine.²⁰ Altogether, it is clear that defective complex I significantly contributes to the observed phenotypes.

PINK1- and Parkin-deficient flies display enlarged, swollen mitochondria, suggesting problems with mitochondrial dynamics and remodeling. These enlarged mitochondria can be rescued via the expression of Drp1 to stimulate fission,^{69–70} which can partially be explained by the fact that Drp1 is phosphorylated in a PINK1-dependent fashion.⁷¹ Recent studies in *Pink1* knock out mice confirmed that Drp1 phosphorylation rescues the *Pink1*-dependent phenotypes.⁷² Furthermore, PINK1-dependent patient-derived fibroblasts show reduced Drp1 phosphorylation.⁷¹ Nonetheless, studies in flies show that the defective complex I function following loss of PINK1 cannot be rescued by expression of Drp1.⁷³ Furthermore, the overexpression of Drp1 causes fragmentation of mitochondria that enables mitochondrial

degradation and turnover, a process in which PINK1 and Parkin operate together.^{69–70,74–75} Interestingly, mitophagy is initially not affected in *Pink1*- and *Parkin*-deficient flies and only declines in aged flies,^{76–77} suggesting that PINK1-Parkin-dependent mitophagy does not play a critical role in the initial stages of PD.

Ceramides and sphingolipids in mitochondrial function

Ceramide can be generated in the mitochondria via CerS or neutral sphingomyelinase^{34–35} and exerts a local function within the mitochondria. Several reports suggest that accumulation of mitochondrial sphingolipids can impair mitochondrial function.^{78–80} Ceramide negatively affects the ETC and its upstream components in a direct and indirect fashion; however, the exact mechanisms remain to be understood. C16-ceramide appears to be the key regulatory ceramide species. C16-ceramide lowers complex I efficiency.⁸¹ Furthermore, overexpression of CerS6, that generates C16-ceramide, inhibits complex II.⁸² In contrast, mice ablated for CerS6 display increased ADP-stimulated complex III activity.⁸³ Interestingly, loss of CerS2 that normally generates very long-chain ceramides, provokes the accumulation of C16-ceramide as a compensatory mechanism and thereby inhibits complex IV activity,⁸⁴ further underlining the importance of C16-ceramide as a regulatory sphingolipid for mitochondrial ETC function and highlighting that dihydroceramide, the precursor of ceramide, does not affect mitochondrial function. Nonetheless, short-chain ceramide also inhibits ETC function and complex I activity.⁸¹ The negative effect of sphingolipids on ETC function can provide an explanation for the observed lower ATP levels in *parkin*-mutant flies as Parkin does not directly affect ETC function. Inhibition of ceramide synthesis, to decrease ceramide levels, improves ATP levels of *Parkin*-deficient flies,²⁰ suggesting that higher ceramide levels cause the lower ATP levels. Finally, C16-ceramide inhibits β -oxidation, as is also seen in heterozygous CerS2 mice.⁸² Interestingly, loss of PINK1 in fibroblasts and flies causes ceramide accumulation that coincides with impaired mitochondrial function and β -oxidation.²⁰ Lowering ceramide levels in both PINK1-deficient flies and patient-derived fibroblasts increases β -oxidation that augments ETC function,^{20,85} suggesting a central role for ceramide in the pathogenesis of PINK1- and Parkin-dependent PD.^{85–86}

While C16-ceramide appears to be the key regulatory sphingolipid for ETC function, CerS1-generated C18-ceramide induces mitophagy. Newly translated CerS1 is transported from the ER to the OMM through mitochondria-associated membranes (MAM),⁸⁷ regulating ceramide-induced mitophagy. Within mitochondria, ceramide induces the formation of activated and lipidated LC3-II, to which it binds with a higher affinity than

to the non-lipidated LC3-I form.⁸⁸ The binding occurs on the mitochondrial membrane and enables the targeting of the mitochondria to the autophagosomes, thus promoting ceramide-induced mitophagy.⁴³ Studies in flies show that following loss of Pink1, ceramide localizes to LC3-II,²⁰ suggesting the observed ceramide accumulation promotes ceramide-induced mitophagy. This provides an explanation for the initial lack of impairment of mitophagy in Pink1- and Parkin-deficient flies that only occurs in aged flies.^{76–77} Interestingly, knock-down of Drp1 causes elongated mitochondria and prevents mitophagy through an alteration of ceramide distribution in mitochondria.^{43,89–90} Furthermore, sphingolipids also play a role in mitochondrial dynamics via the binding of C16-ceramide generated by CerS6, but not by CerS5, to Mfn.⁸³ Thus, sphingolipids directly affect those aspects of mitochondrial function that are also affected in PD, including ETC function, mitochondrial remodelling and mitophagy. Furthermore, a direct connection was identified as ceramide is elevated as a response on loss of PINK1, further aggravating the phenotypes.²⁰ In addition, these data directly link mitophagy and ETC defects and thus highlight the importance of ceramide in the pathogenesis of PD and its mitochondrial defects.

Ceramides and Sphingolipids in Relation to Autophagy

Autophagy

Autophagy is a tightly regulated, highly conserved process in which the cell self-digests to get rid of damaged organelles, misfolded proteins or microorganisms, and can either be pro-survival or end up in cell death.³⁸ Autophagy can be subdivided into three different forms, namely macroautophagy, microautophagy, and chaperone-mediated autophagy.⁹¹ We will mostly discuss macroautophagy that we will refer to as autophagy hereafter.

Autophagy entails five steps, in which more than 30 autophagy-related proteins (Atgs) are involved⁹² (Figure 3). The first step of autophagy is initiation, in which a double membrane structure is formed. Initiation of autophagy is kept at low levels without a stimulus that is regulated by the mammalian target of rapamycin (mTOR) system.⁹³ Upon starvation, AMP-activated protein kinase (AMPK) is activated to phosphorylate and thereby inhibit mTOR, followed by the initiation of autophagy^{94–95} and thus, starvation-mediated autophagy provides nutrients to the cell by recycling existing contents.⁹³ Secondly, phagophore nucleation is regulated by the Beclin 1-containing phosphatidylinositol 3-kinase (PI3K) complex.^{96–97} Phagophore elongation is the third step that exists of two ubiquitin-like reactions, in which Atg12 ubiquitinates Atg5 and LC3 that consists of two forms (LC3-I and LC3-II). The inactive LC3-I covalently binds to

phosphatidylethanolamine (PE) resulting in the activated form, LC3-II.^{92–93,98} This binding is dependent on Atg3 and Atg7.^{92–93} Loss of Atg3 impairs this binding and therefore blocks the activation of LC3 to induce autophagosome formation.⁹⁹ The mature autophagosome is formed following phagophore closure that entails the fourth step in autophagy.⁹² The fifth and final step of autophagy is the fusion of the autophagosome to the lysosome, either directly or via an in-between fusion to the endosomes prior to fusing to the lysosomes, to form autolysosomes.⁹³ This fusion is regulated by a multi-protein complex that includes Rab GTPases, soluble N-ethylmaleimide-sensing factor attachment protein receptors (SNAREs), membrane tethering complexes such as the HOPS complex and dynein- and kinesin-dependent transport along the microtubules.^{100–103} In this complex, Rab7 aids in the transport along the microtubule to the lysosomes.^{101–106} Subsequently, organelles, proteins etc. will be degraded in the lysosomes.⁹² Lysosomes are membrane-bound organelles that contain an acidic lumen to enable the activity of hydrolytic enzymes to break down biomolecules which constitutes the end-stage of autophagy. A dysfunctional lysosome will impair autophagy flux.

Mitophagy is a specific form of autophagy that clears defective mitochondria. Mitophagy can be regulated via different mechanisms. Following a defective mitochondrial membrane potential, PINK1 remains uncleaved on the outer mitochondrial membrane, providing a signal to Parkin to relocate to the mitochondria where it can ubiquitinate OMM proteins to label it for LC3-mediated degradation.¹⁰⁷ Alternatively, mitophagy can be regulated via ceramide accumulation at the mitochondria (Figure 2).⁴³

Role of autophagy in PD

Impaired autophagy in a tissue-specific Atg7 knockout mice leads to age-dependent loss of dopaminergic neurons,¹⁰⁸ implicating autophagy in the pathogenesis of PD. Postmortem brains of PD patients demonstrate anomalies in autophagy in dopaminergic neurons.¹⁰⁹ Furthermore, LC3 was present in the Lewy bodies examined, and increases in the lysosomal markers of autophagy precede these Lewy body depositions in brains of PD patients.^{110–112} The pathological hallmark of PD is the existence of Lewy bodies that mostly contain aggregations of alpha-synuclein alongside lipids, membrane fragments and disrupted organelles, suggesting that an impairment in the removal of alpha-synuclein via autophagy is key to the formation of these depositions. Furthermore, loss of cathepsin-D (CTSD), a lysosomal hydrolase, leads to oligomeric alpha-synuclein aggregates in dopaminergic cells,^{113–114} suggesting CTSD plays a crucial role in the breakdown of alpha-synuclein. Interestingly, mutations in *VPS35* cause cellular mislocalization of CTSD in *Drosophila*, leading to

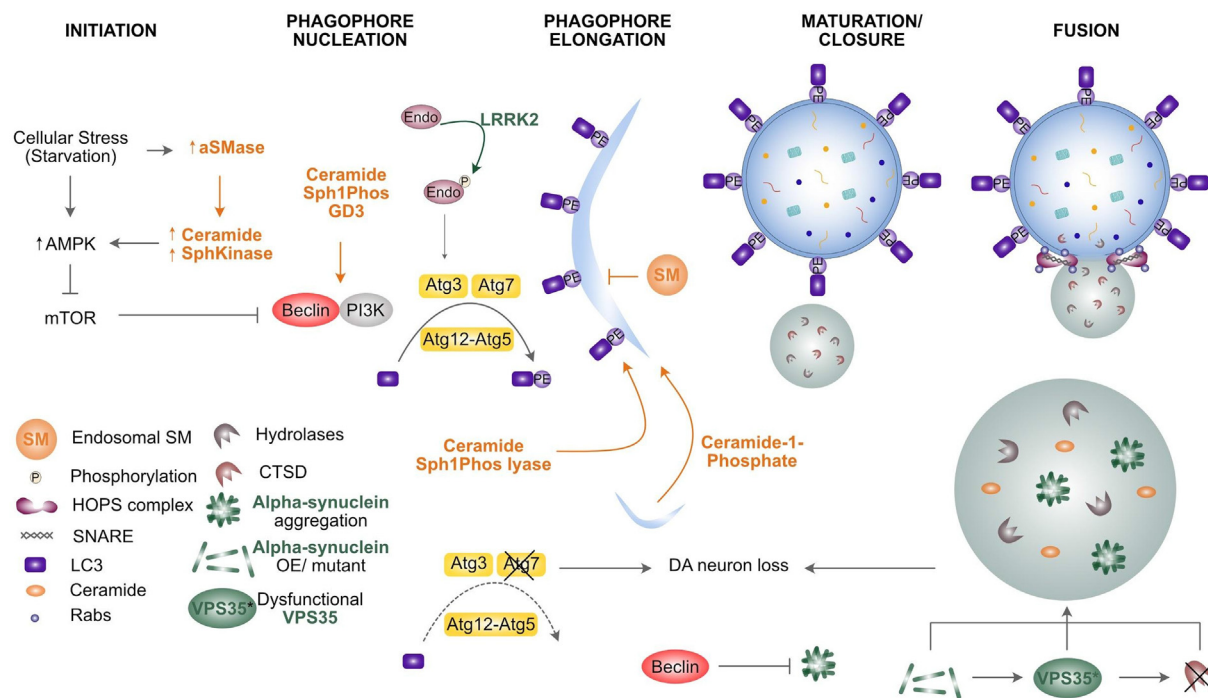


Figure 3. A simplified scheme of the implications of PD and sphingolipids on autophagy. Cellular stress activates AMPK, inhibiting mTOR's inhibiting function on autophagy initiation. Different sphingolipids influence these initial steps of the initiation and phagophore nucleation. Beclin prevents alpha-synuclein aggregation. The lipidation of LC3 activates LC3, which is partially under the control of LRRK2-dependent phosphorylation of Endophilin (Endo). Loss of Atg7 prevents the activation of LC3, which results in dopaminergic (DA) neuron loss. Phagophore elongation depends on additionally presented membranes activated by ceramide sphingosine-1-phosphate lyase and ceramide-1-phosphate and inhibited by sphingomyelin (SM). Following the maturation and closure, the autophagosome will fuse with the lysosomes, facilitated by a complex comprising SNARE, HOPS, and Rab proteins. The fusion is inhibited by overexpression (or mutant) alpha-synuclein, dysfunctional VPS35, and lack of CTSD. Overexpression of alpha-synuclein inhibits functional VPS35 that further mislocalizes CTSD. In addition, the lack of this fusion will result in increased ceramide and enlarged lysosomes. PD-related genes are indicated in green. PI3K: phosphatidylinositol 3-kinase; GD3: ganglioside; AMPK: AMP-activated protein kinase; mTOR: mammalian target of rapamycin; SNARE: soluble N-ethylmaleimide-sensing factor attachment protein receptors; CTSD: cathepsin D; Sph1Phos lyase: sphingosine 1-phosphate.

further accumulation of alpha-synuclein¹¹³ and indicating a close interplay between alpha-synuclein and VPS35. In addition, *VPS35* and *SNCA* mutations display enlarged lysosomes in flies and cells, respectively,²¹ supporting the notion that a defective autophagy process plays an essential role in PD. Furthermore, Beclin-1, that regulates phagophore nucleation in autophagy, lowers alpha-synuclein aggregation in neuronal cells when overexpressed.¹¹⁵

Mutations in *LRRK2*, the most common genetic cause of PD, are also observed in sporadic forms of PD, and fibroblasts isolated from such patients display altered ratios of lipidated activated LC3-II over the inactive LC3-I.¹¹⁶ In addition, *LRRK2* accumulates in Rab7- and LAMP2-positive granules in post-mortem brain samples of patients suffering from sporadic PD, and rodent models with *LRRK2* knockout display an age-dependent elevation of lysosomal proteins,¹¹⁷ further demonstrating the

importance of autophagy in genetic and sporadic forms of PD. In addition, studies in flies show that *LRRK2*-mediated phosphorylation of endophilin A recruits Atg3 to the synapse where it provokes autophagy.^{118–119} While the kinase activity of *LRRK2* is thus relevant for the observed autophagy-related defects, it must be remembered that PD-causing mutations in *LRRK2* are not limited to the kinase domain, and are found throughout the entire gene.¹²⁰

Role of Ceramides/Sphingolipids in autophagy

Autophagy entails several steps in which all sphingolipids play a crucial role. Induction of autophagy often occurs via starvation that inactivates mTOR^{94–95} that is sphingolipid dependent. Starvation induces activation of acid sphingomyelinase that increases ceramide and sphingosine kinase,¹²¹ while loss of acid

sphingomyelinase or sphingosine kinase 1 disrupts starvation-induced autophagy.^{121–123} Interestingly, exogenous ceramide promotes autophagy by mimicking starvation.¹²⁴ The Beclin-PI3K complex that regulates the phagophore nucleation is under the control of a combination of sphingolipids, including ceramide, sphingosine-1-phosphate and the ganglioside GD3.^{125–126} Via this regulation of Beclin-1, these sphingolipids, we believe, may be able to indirectly affect cellular amounts of aggregated alpha-synuclein. During phagophore elongation, LC3-II activation is induced by ceramide and sphingosine-1-phosphate lyase 1.⁹² Furthermore, elongation requires vesicles to provide membranes, which is promoted by ceramide-1-phosphate and inhibited by sphingomyelin accumulation in the endosomes.⁹² Remarkably, mitochondrial ceramide was found to directly bind to lipidated LC3-II,⁴³ which was also observed in *pink1*-mutant flies in which ceramide-induced mitophagy was activated.²⁰ As mentioned above, activation of acid sphingomyelinase increases ceramide levels, which can impair the trafficking of the autophagosomes along the microtubules via the acetylation of tubulin.¹²⁷ Furthermore, sphingosine promotes SNARE-mediated membrane fusion during neuronal exocytosis.⁹² Whether this function also affects autophagosome-lysosomal fusion needs to be further elucidated.

In VPS35 and alpha-synuclein fly models of PD, enlarged lysosomes are observed, suggesting that both lysosomal function and autophagy flux is impaired. Interestingly, these PD models have increased ceramide levels that appear to be trapped in the lysosomes, and this effect can be rescued by lowering ceramide levels.²¹ Furthermore, key proteins in the autophagy process, such as LC3 and LAMP1 or LAMP2, show abnormalities in PD, while under regulation of specific sphingolipids, further highlighting the important interaction between sphingolipids and autophagy in PD.

Ceramides and Sphingolipids in Relation to Endosomal Trafficking

Endosomal trafficking

Endocytic vesicle recycling is a highly conserved system that maintains the turnover and ensures quality of cellular components. It delivers receptors, molecules, proteins, and lipids to subcellular organelles to be recycled or to lysosomes to be degraded.^{128–130} This protein sorting system consists of two major components: the endosomes and the retromer complex (Figure 4). Endosomes can be subdivided into early endosomes (EE), recycling endosomes (RE), and late endosomes (LE), also termed multivesicular bodies (MVBs).^{131–132} At the same time, the retromer complex is a trimer complex of VPS26, VPS29, and VPS35.^{133–135} Furthermore, endosomal trafficking

is supported by the Rab GTPase protein family, which exerts many functions dependent on its members' cellular localization.^{136–137}

The early endosome is membrane-bound, dynamic, and contains membrane-bound cargo as a consequence of clathrin-mediated endocytosis from the cell surface, or via anterograde transport from the Golgi network.^{138–139} Often, the internalization of plasma membrane-bound receptors is a means to regulate signaling.^{140–141} The resulting vesicles undergo recycling directly to the membrane or via the RE, from where they will be sorted back to the membrane. The fusion to the RE depends on Rab5.¹⁴² As mentioned above, the presentation of proteins to the lysosome can occur indirectly via the endosome in which the EE with the incorporated proteins mature into the LE.^{143–144} For degradation via the endo-lysosome, proteins are tagged with a polyubiquitin tail that is recognized by the endosomal-sorting-complexes-required-for-transport complex (ESCRT) which sorts them into intra-endosomal vesicles resulting in the formation of MVBs.^{145–146} Subsequently, the MVBs fuse with the lysosomes, which is mediated by Rab7.¹⁴²

The cargo sorting is tightly balanced to ensure three different outcomes: the recycling of cargo from the endosome to the plasma membrane; the retrograde transport to the trans-Golgi network (TGN); the degradation of cargo via EE into the lysosome.^{138–139} The retromer is an essential complex in the protein sorting system from the endosomes to the TGN.^{133,147–148} The trimer of core proteins (VPS26, VPS29, and VPS35) is responsible for the recognition and binding of the cargo, and forms a complex with a dimer of sorting nexin, that then sorts the cargo for its transport to the TGN.^{133,139,149} The retromer is localized on endosomes in a Rab7-dependent fashion.^{136,150–153} Inactivated Rab7a lowers VPS35 on the endosomes and therefore impairing retromer function.^{153–156}

Endosomal trafficking defects in PD

A variety of Rab proteins interact with multiple PD-linked proteins with a role in normal endosomal trafficking. Alpha-synuclein interacts with multiple Rab-proteins and can impair several steps in the endo-lysosomal pathway, including ER-Golgi trafficking.^{157–158} Overexpression of Rab1 improves dopaminergic neuron loss in alpha-synuclein animal models,¹⁵⁸ supporting the importance of alpha-synuclein in endosomal mechanisms. In addition, alpha-synuclein overexpression in mammalian cells increases golgi fragmentation that can then be rescued by overexpression of Rab1.¹⁵⁹ Furthermore, alpha-synuclein interacts with Rab8, which is involved in post-Golgi vesicle trafficking, to reduce cellular toxicity.¹⁶⁰ Conversely, Rab7, which mediates fusion of the autophagosome or MVB to the lysosome, clears alpha-synuclein aggregates in HEK293 cells.¹⁶¹ Rab7 and Rab8, in addition to other Rab proteins,

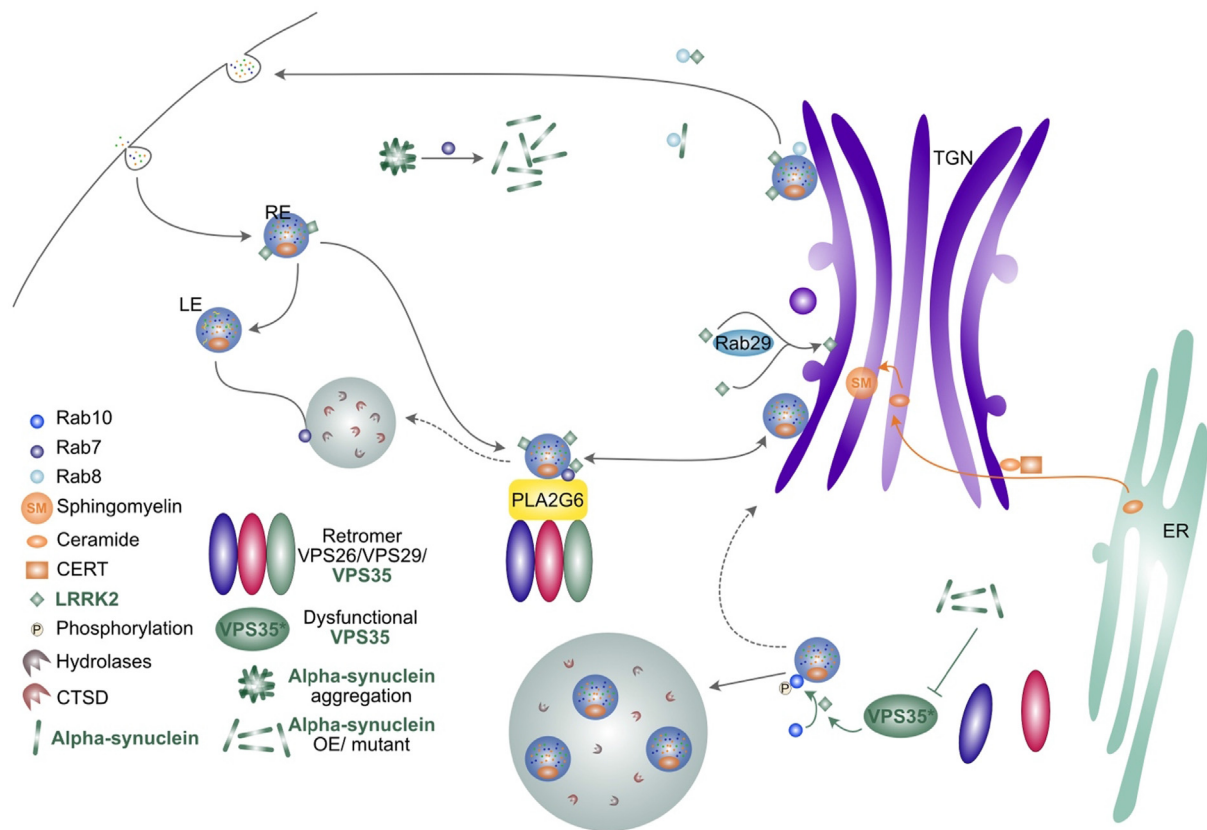


Figure 4. A simplified scheme of the implications of PD and sphingolipids on endosomal trafficking.

Vesicles are recycled between the plasma membrane and the TGN and ER. RE can transform into LE that fuse with the lysosomes for degradation or binds to the retromer. LRRK2 will bind to the endosomes and their accompanying Rab proteins. Subsequently, the endosomes are transported to the TGN. LRRK2 will be translocated to the TGN, regulated by Rab29, and further recruits Rab proteins. A small subset of the endosomes will fuse with the lysosomes. Endosomes will be recycled with newly formed vesicles and finally be transported to the plasma membrane. Rab7 dissolves the alpha-synuclein aggregates. ER-generated ceramide will be transported to the TGN via the CERT, where it will be converted to SM. Overexpression of alpha-synuclein will inhibit retromer function and will prevent endosomes from being translocated to the TGN, allowing them to fuse to the lysosomes, which then become enlarged. PD-related genes are indicated in green. CERT: ceramide transporter protein; ER endoplasmic reticulum; TGN: trans Golgi network; RE: recycling endosome; LE: late endosome.

also interact with LRRK2,^{162–163} which is localized on endosomes. Ablation of LRRK2 causes fragmentation of the Golgi,¹⁶⁴ similar to alpha-synuclein overexpression. Regulation of LRRK2 occurs via Rab29, which recruits LRRK2 to the TGN, where it prevents fragmentation of the Golgi and lysosomes. In addition, Rab29 activates LRRK2 kinase activity that further recruits additional Rab proteins.^{165–170} Interestingly, the PD-causing mutation of VPS35 increases LRRK2-dependent phosphorylation of Rab10 on the endosomes in mice,¹⁷¹ supporting the role of VPS35 in regulating LRRK2 activity at the endosomes. Furthermore, stimulation of retromer function alleviates the enlarged endosomes and autophagosomes observed in LRRK mutant *Drosophila melanogaster*, and rescues the observed sorting defects and neurite shortening in LRRK2-related PD models.¹⁷²

VPS35, mutations in which cause PD, encodes VPS35, a protein of the core trimer of the retromer complex (PD), and thus provides the most obvious relationship between endosomal trafficking abnormalities and PD. The importance of VPS35 in endocytosis is further demonstrated by studies in fruit flies where loss of Vps35 blocks scavenger receptor ligand endocytosis and results into several endocytic proteins to be mislocalized.¹⁷³ Furthermore, reduced VPS35 protein levels were observed in the *substantia nigra* of post-mortem brain of patients suffering from PD.¹⁶⁵ In addition, the overexpression of alpha-synuclein decreases VPS35 and VPS26 protein levels in human cells,²¹ suggesting a negative correlation between alpha-synuclein and VPS35. While the complete mechanism remains elusive, compelling data indicate the importance of endosomal trafficking defects in the pathogenesis of PD.

Role of ceramide/sphingolipids in endosomal trafficking

Sphingolipids are transported via the endosomal trafficking system to the correct subcellular compartments to execute their function. *De novo* synthesized ceramide will be transported from the ER to the TGN via CERT to synthesize sphingomyelin, or to the *cis*-Golgi pole via vesicle-mediated transport to generate glucosylceramide (GlcCer).^{174–175} This GlcCer will be either converted to glycosphingolipids as they are processed through Golgi cisternae, or be transported to the TGN via the GlcCer transfer protein.^{176–178} In the TGN, sphingolipids will be packaged into membrane carriers, followed by their transport to the appropriate subcellular compartments, including the plasma membrane.^{179–181} Conversely, sphingolipids are recycled, for which they undergo retromer-mediated transport from the plasma membrane to the TGN via endocytic vesicles.^{178,182} A small subset is transported to the lysosomes for degradation to ceramide via acid sphingomyelinase.^{21,182} Interestingly, studies show that the fly ortholog of the phospholipase PLA2G6 regulates retromer function and thus also sphingolipid transport. Furthermore, a defective or inactive retromer function as a consequence of *PLA2G6* mutations, *VPS35* mutations, or alpha-synuclein overexpression, increases the transport of sphingomyelin to the lysosomes thus resulting in increased ceramide levels that reside in enlarged lysosomes.²¹ *PLA2G6* is also a parkinsonism-associated gene, and thus the retromer that is critical for proper sphingolipid metabolism partly consists of, or is regulated by, genes that are genetically linked to PD. Finally, mutations in *GBA*, which converts the GlcCer to ceramide, which is then transported to the Golgi, are also strongly associated with PD. Thus, several PD-linked genes play a crucial role in the transport and metabolism of sphingolipids.

Conclusion

Sphingolipids, and more specifically ceramide itself, play a crucial role in several of the cellular processes known to be involved in PD. However, in PD patients, data on sphingolipids are seemingly contradictory. Our current knowledge shows that these discrepancies are strongly dependent on which tissue is investigated. Therefore, the exact mechanisms, and sphingolipid species, that appear to be the most relevant for PD need further investigation in the relevant tissues.

In-depth knowledge can potentially allow sphingolipids to serve as biomarkers or, more excitingly, may provide novel therapeutic targets. Several drugs to manipulate sphingolipid homeostasis are currently under investigation in clinical trials, such as myriocin for cancer, or miglustat for neurological disorders such as

Niemann-Pick Type C disease. Sphingolipids are essential lipids in almost all cellular processes; hence their manipulation may yield unwanted side effects. For instance, elevated ceramide levels inhibit β -oxidation and, therefore, negatively impact the efficiency of the ETC, as was shown recently in *PINK1*-deficient flies and fibroblasts.²⁰ Interestingly, stimulation of β -oxidation, via the supplementation of carnitine, alleviates *PINK1*-mutant phenotypes,²⁰ providing an alternative strategy for treating *PINK1*-related PD.

Population genetic studies have revealed a wide variety in the circulating levels of Cer/SL between individuals and have identified the major human genes involved in sphingolipid metabolism that regulate these levels.¹⁸³ Given the critical regulatory role played by some of these lipid species, further studies will be warranted to indicate whether this variation plays a role in the manifestation of the disease in relation to the observed differences in the age of onset or rate of disease progression. In addition, it remains unclear if the observed alterations in sphingolipids are the cause or the result of the disease. Our recent data showing that increased ceramide is adding on to the phenotypes,²⁰ suggest that it is at least in part the cause of the phenotypes. However, additional longitudinal epidemiological studies in which sphingolipids or their genetic determinants are monitored will better address this question.

Finally, sphingolipids are not limited to playing a role only in the pathogenesis of PD: an increasing amount of evidence shows that sphingolipids are involved in other neurodegenerative disorders, including Alzheimer's disease or Fatty acid hydroxylase-associated neurodegeneration and other neurodegeneration with brain iron stimulation.^{86,184} Given what we know so far, it is timely to understand not only the roles that sphingolipids may play in disease pathogenesis in the brain, but the extent to which they do. This can be achieved either by targeted pharmaceutical, dietary or genetic intervention, where concentrations of key modulatory sphingolipids may be manipulated to yield meaningful neuroprotective effects.

CRedit authorship contribution statement

Melissa Vos: Conceptualization, Writing – original draft, Writing – review & editing, Visualization. **Christine Klein:** Conceptualization, Writing – review & editing. **Andrew A Hicks:** Conceptualization, Writing – original draft, Writing – review & editing.

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Declaration of Competing Interest

CK is a medical advisor to Centogene for genetic testing in the field of movement disorders and dementia, excluding Parkinson's disease, received speakers' honoraria from Desitin and Bial, and is a member of the scientific advisory board of Retromer Therapeutics.

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References

- Tysnes, O.-B., Storstein, A., (2017). Epidemiology of Parkinson's disease. *J. Neural Transm.* **124**, 901–905. <https://doi.org/10.1007/s00702-017-1686-y>.
- Savica, R., Grossardt, B.R., Bower, J.H., Eric Ahlskog, J., Rocca, W.A., (2016). Time Trends in the Incidence of Parkinson Disease. *JAMA Neurol.* **73**, 981–989. <https://doi.org/10.1001/JAMANEUROL.2016.0947>.
- Ou, Z., Pan, J., Tang, S., Duan, D., Yu, D., Nong, H., Wang, Z., (2021). Global Trends in the Incidence, Prevalence, and Years Lived With Disability of Parkinson's Disease in 204 Countries/Territories From 1990 to 2019. *Front. Public Health* **9** <https://doi.org/10.3389/FPUH.2021.776847>.
- Dorsey, E.R., Sherer, T., Okun, M.S., Bloem, B.R., (2018). The Emerging Evidence of the Parkinson Pandemic. *J. Parkinsons Dis.* **8**, S3–S8. <https://doi.org/10.3233/JPD-181474>.
- Bloem, B.R., Okun, M.S., Klein, C., (2021). Parkinson's disease. *Lancet* **397**, 2284–2303.
- Marras, C., Lang, A., van de Warrenburg, B., Sue, C., Tabrizi, S., Bertram, L., Mercimek-Mahmutoglu, S., Ebrahimi-Fakhari, D., et al., (2016). Nomenclature of genetic movement disorders: Recommendations of the international Parkinson and movement disorder society task force. *Mov. Disord.* **31**, 436–457.
- Shahmoradian, S.H., Lewis, A.J., Genoud, C., Hench, J., Moors, T.E., Navarro, P.P., Castaño-Díez, D., Schweighauser, G., et al., (2019). Lewy pathology in Parkinson's disease consists of crowded organelles and lipid membranes. *Nature Neurosci.* **22**, 1099–1109. <https://doi.org/10.1038/s41593-019-0423-2>.
- Kasten, M., Hartmann, C., Hampf, J., Schaake, S., Westenberger, A., Vollstedt, E.J., Balck, A., Domingo, A., et al., (2018). Genotype-Phenotype Relations for the Parkinson's Disease Genes Parkin, PINK1, DJ1: MDSGene Systematic Review. *Mov. Disord.* <https://doi.org/10.1002/mds.27352>.
- Chaudhuri, K.R., Healy, D.G., Schapira, A.H., (2006). Non-motor symptoms of Parkinson's disease: diagnosis and management. *Lancet Neurol.* **5**, 235–245 http://www.ncbi.nlm.nih.gov/entrez/query.fcgi?cmd=Retrieve&db=PubMed&dopt=Citation&list_uids=16488379.
- Lee, H., Koh, S., (2015). Many Faces of Parkinson's Disease: Non-Motor Symptoms of Parkinson's Disease. *Mov. Disord.* **8**, 92–97.
- Chaudhuri, K.R., Schapira, A.H., Jenner, P., (2017). Non-motor features of Parkinson disease. *Nature Rev. Neurosci.* **18**, 435. <https://doi.org/10.1038/nrn.2017.62>.
- Kasten, M., Marras, C., Klein, C., (2017). Nonmotor Signs in Genetic Forms of Parkinson's Disease, in: *Int. Rev. Neurobiol.*, 129–178. <https://doi.org/10.1016/bs.irn.2017.05.030>.
- You, H., Mariani, L., Mangone, G., le Febvre, D., de Nailly, F., Charbonnier-Beaupel, J.C., (2018). Molecular basis of dopamine replacement therapy and its side effects in Parkinson's disease. *Cell Tissue Res. Cell Tissue Res.* **373**, 111–135.
- Vos, M., Klein, C., (2021). The Importance of Drosophila melanogaster Research to UnCover Cellular Pathways Underlying Parkinson's Disease. *Cells*. **10**, 579. <https://doi.org/10.3390/cells10030579>.
- Ivatt, R., Sanchez-Martinez, A., Godena, V., Brown, S., Ziviani, E., Whitworth, A., (2014). Genome-wide RNAi screen identifies the Parkinson disease GWAS risk locus SREBF1 as a regulator of mitophagy. *PNAS* **111**, 8494–8499.
- Vos, M., Geens, A., Böhm, C., Deaulmerie, L., Swerts, J., Rossi, M., Craessaerts, K., Leites, E.P., et al., (2017). Cardiolipin promotes electron transport between ubiquinone and complex I to rescue PINK1 deficiency. *J. Cell Biol.* **216**, 695–708.
- Zanon, A., Kalvakuri, S., Rakovic, A., Foco, L., Guida, M., Schwenbacher, C., Serafin, A., Rudolph, F., et al., (2017). SLP-2 interacts with Parkin in mitochondria and prevents mitochondrial dysfunction in Parkin-deficient human iPSC-derived neurons and Drosophila. *Hum. Mol. Genet.* epub ahead.
- Mielke, M.M., Maetzler, W., Haughey, N.J., Bandaru, V.V., Savica, R., Deuschle, C., Gasser, T., Hauser, A.-K., et al., (2013). Plasma Ceramide and Glucosylceramide Metabolism Is Altered in Sporadic Parkinson's Disease and Associated with Cognitive Impairment: A Pilot Study. *PLoS One* **8**, e73094.
- Abbott, S., Li, H., Muñoz, S., Knoch, B., Batterham, M., Murphy, K., Halliday, G., Garner, B., (2014). Altered ceramide acyl chain length and ceramide synthase gene expression in Parkinson's disease. *Mov. Disord.* **29**, 518–526.
- Vos, M., Dulovic-Mahlow, M., Mandik, F., Frese, L., Kanana, Y., Diaw, S.H., Depperschmidt, J., Böhm, C., et al., (2021). Ceramide accumulation induces mitophagy and impairs β -oxidation in PINK1 deficiency. *Proc. Natl. Acad. Sci.* **118** <https://doi.org/10.1073/PNAS.2025347118>. e2025347118.

21. Lin, G., Lee, P.-T., Chen, K., Mao, D., Tan, K.L., Zuo, Z., Lin, W.-W., Wang, L., et al., (2018). Phospholipase PLA2G6, a Parkinsonism-Associated Gene, Affects Vps26 and Vps35, Retromer Function, and Ceramide Levels, Similar to α -Synuclein Gain. *Cell Metab.* **28**, 605–618.e6. <https://doi.org/10.1016/j.cmet.2018.05.019>.
22. Goya, M.E., Xue, F., Sampedro-Torres-Quevedo, C., Arnaouteli, S., Riquelme-Dominguez, L., Romanowski, A., Brydon, J., Ball, K.L., et al., (2020). Probiotic *Bacillus subtilis* Protects against α -Synuclein Aggregation in *C. elegans*. *Cell Rep.* **30**, 367–380.e7. <https://doi.org/10.1016/j.celrep.2019.12.078>.
23. Hannun, Y.A., Obeid, L.M., (2008). Principles of bioactive lipid signalling: Lessons from sphingolipids. *Nature Rev. Mol. Cell Biol.* **9**, 139–150. <https://doi.org/10.1038/nrm2329>.
24. Futerman, A.H., Hannun, Y.A., (2004). The complex life of simple sphingolipids. *EMBO Rep.* **5**, 777–782. <https://doi.org/10.1038/sj.embor.7400208>.
25. Merrill, A.H., (2002). De novo sphingolipid biosynthesis: A necessary, but dangerous, pathway. *J. Biol. Chem.* **277**, 25843–25846. <https://doi.org/10.1074/jbc.R200009200>.
26. Kogot-Levin, A., Saada, A., (2014). Ceramide and the mitochondrial respiratory chain. *Biochimie* **100**, 88–94. <https://doi.org/10.1016/j.biochi.2013.07.027>.
27. Hornemann, T., Penno, A., Rütli, M.F., Ernst, D., Kivrak-Pfiffner, F., Rohrer, L., von Eckardstein, A., (2009). The SPTLC3 subunit of serine palmitoyltransferase generates short chain sphingoid bases. *J. Biol. Chem.* **284**, 26322–26330. <https://doi.org/10.1074/JBC.M109.023192>.
28. Harmon, J.M., Bacikova, D., Gable, K., Gupta, S.D., Han, G., Sengupta, N., Somashekarappa, N., Dunn, T.M., (2013). Topological and functional characterization of the ssSPTs, small activating subunits of serine palmitoyltransferase. *J. Biol. Chem.* **288**, 10144–10153. <https://doi.org/10.1074/JBC.M113.451526>.
29. Kitatani, K., Idkowiak-Baldys, J., Hannun, Y.A., (2008). The sphingolipid salvage pathway in ceramide metabolism and signaling. *Cell. Signal.* **20**, 1010–1018. <https://doi.org/10.1016/J.CELLSIG.2007.12.006>.
30. Burger, K.N.J., van der Bijl, P., van Meer, G., (1996). Topology of sphingolipid galactosyltransferases in ER and Golgi: transbilayer movement of monohexosyl sphingolipids is required for higher glycosphingolipid biosynthesis. *J. Cell Biol.* **133**, 15–28. <https://doi.org/10.1083/JCB.133.1.15>.
31. Andrieu-Abadie, N., Levade, T., (2002). Sphingomyelin hydrolysis during apoptosis. *BBA* **1585**, 126–134. [https://doi.org/10.1016/S1388-1981\(02\)00332-3](https://doi.org/10.1016/S1388-1981(02)00332-3).
32. Kumagai, K., Kawano-Kawada, M., Hanada, K., (2014). Phosphoregulation of the ceramide transport protein CERT at serine 315 in the interaction with VAMP-associated protein (VAP) for inter-organellar trafficking of ceramide in mammalian cells. *J. Biol. Chem.* **289**, 10748–10760. <https://doi.org/10.1074/JBC.M113.528380>.
33. Kumagai, K., Yasuda, S., Okemoto, K., Nishijima, M., Kobayashi, S., Hanada, K., (2005). CERT Mediates Intermembrane Transfer of Various Molecular Species of Ceramides. *J. Biol. Chem.* **280**, 6488–6495. <https://doi.org/10.1074/JBC.M409290200>.
34. Rajagopalan, V., Canals, D., Luberto, C., Snider, J., Voelkel-Johnson, C., Obeid, L.M., Hannun, Y.A., (1950). Critical determinants of mitochondria-associated neutral sphingomyelinase (MA-nSMase) for mitochondrial localization. *BBA* **2015**, 628–639. <https://doi.org/10.1016/J.BBAGEN.2014.11.019>.
35. Hannun, Y.A., Obeid, L.M., (2018). Sphingolipids and their metabolism in physiology and disease. *Nature Rev. Mol. Cell Biol.* **19**, 175–191. <https://doi.org/10.1038/NRM.2017.107>.
36. Ogretmen, B., Hannun, Y.A., (2004). Biologically active sphingolipids in cancer pathogenesis and treatment. *Nature Rev. Cancer* **4**, 604–616. <https://doi.org/10.1038/nrc1411>.
37. Mizutani, Y., Kihara, A., Igarashi, Y., (2005). Mammalian Lass6 and its related family members regulate synthesis of specific ceramides. *Biochem. J* **390**, 263–271. <https://doi.org/10.1042/BJ20050291>.
38. Jiang, W., Ogretmen, B., (2014). Autophagy paradox and ceramide. *Biochim. Biophys. Acta (BBA) – Mol. Cell Biol. Lipids* **1841**, 783–792. <https://doi.org/10.1016/J.BBALIP.2013.09.005>.
39. Saddoughi, S.A., Ogretmen, B., (2013). Diverse Functions of Ceramide in Cancer Cell Death and Proliferation. *Adv. Cancer Res.* **117**, 37–58. <https://doi.org/10.1016/B978-0-12-394274-6.00002-9>.
40. Saddoughi, S.A., Song, P., Ogretmen, B., (2008). Roles of bioactive sphingolipids in cancer biology and therapeutics. *Subcell. Biochem.* **49**, 413–440. https://doi.org/10.1007/978-1-4020-8831-5_16/COVER.
41. Dany, M., Ogretmen, B., (1853). Ceramide induced mitophagy and tumor suppression. *BBA* **2015**, 2834–2845. <https://doi.org/10.1016/J.BBAMCR.2014.12.039>.
42. Wittenbecher, C., Cuadrat, R., Johnston, L., Eichelmann, F., Jäger, S., Kuxhaus, O., Prada, M., del Greco, F., et al., (2022). Dihydroceramide- and ceramide-profiling provides insights into human cardiometabolic disease etiology. *Nature Commun.* **13**, 1–13. <https://doi.org/10.1038/s41467-022-28496-1>.
43. Sentelle, R.D., Senkal, C.E., Jiang, W., Ponnusamy, S., Gencer, S., Panneer Selvam, S., Ramshesh, V.K., Peterson, Y.K., et al., (2012). Ceramide targets autophagosomes to mitochondria and induces lethal mitophagy. *Nature Chem. Biol.* **8**, 831. <https://doi.org/10.1038/nchembio.1059>.
44. Kwak, S.H., Park, K.S., Lee, K.U., Lee, H.K., (2010). Mitochondrial metabolism and diabetes. *J Diabetes Invest.* **1**, 161–169. <https://doi.org/10.1111/J.2040-1124.2010.00047.X>.
45. Papa, S., Martino, P.L., Capitanio, G., Gaballo, A., de Rasmio, D., Signorile, A., Petruzzella, V., (2012). The oxidative phosphorylation system in mammalian mitochondria. *Adv. Exp. Med. Biol.* **942**, 3–37. https://doi.org/10.1007/978-94-007-2869-1_1.
46. Bartlett, K., Eaton, S., (2004). Mitochondrial beta-oxidation. *Eur. J. Biochem.* **271**, 462–469. <https://doi.org/10.1046/J.1432-1033.2003.03947.X>.
47. Scott, I., Youle, R.J., (2010). Mitochondrial fission and fusion. *Essays Biochem.* **47**, 85–98. <https://doi.org/10.1042/bse0470085>.
48. Youle, R.J., van der Bliek, A.M., (2012). Mitochondrial fission, fusion, and stress. *Science* **337**, 1062–1065. <https://doi.org/10.1126/science.1219855>.
49. Rube, D., van der Bliek, A., (2004). Mitochondrial morphology is dynamic and varied. *Mol. Cell. Biochem.* **256**, 331–339.
50. Vázquez-Trincado, C., García-Carvajal, I., Pennanen, C., Parra, V., Hill, J.A., Rothermel, B.A., Lavandero, S.,

- (2016). Mitochondrial dynamics, mitophagy and cardiovascular disease. *J. Physiol.* **594**, 509–525. <https://doi.org/10.1113/JP271301>.
51. Vos, M., Lauwers, E., Verstreken, P., (2010). Synaptic mitochondria in synaptic transmission and organization of vesicle pools in health and disease. *Front. Synaptic Neurosci.* **2**, 12. <https://doi.org/10.3389/fnsyn.2010.00139>.
 52. Zenisek, D., Matthews, G., (2000). The role of mitochondria in presynaptic calcium handling at a ribbon synapse. *Neuron* **25**, 229–237. [https://doi.org/10.1016/S0896-6273\(00\)80885-5](https://doi.org/10.1016/S0896-6273(00)80885-5).
 53. Duchen, M.R., (1992). Ca(2+)-dependent changes in the mitochondrial energetics in single dissociated mouse sensory neurons. *Biochem. J* **283** (Pt 1), 41–50. <https://doi.org/10.1042/BJ2830041>.
 54. Graier, W.F., Frieden, M., Malli, R., (2007). Mitochondria and Ca(2+) signaling: old guests, new functions. *Pflugers Arch.* **455**, 375–396. <https://doi.org/10.1007/S00424-007-0296-1>.
 55. Gunter, T.E., Sheu, S.S., (2009). Characteristics and possible functions of mitochondrial Ca(2+) transport mechanisms. *BBA* **1787**, 1291–1308. <https://doi.org/10.1016/J.BBABIO.2008.12.011>.
 56. Apicco, D.J., Shlevkov, E., Nezich, C.L., Tran, D.T., Guilmette, E., Nicholatos, J.W., Bantle, C.M., Chen, Y., et al., (2021). The Parkinson's disease-associated gene ITPKB protects against α -synuclein aggregation by regulating ER-to-mitochondria calcium release. *PNAS* **118**. <https://doi.org/10.1073/PNAS.2006476118/-/DCSUPPLEMENTAL>.
 57. Langston, J.W., Ballard, P., Tetrud, J.W., Irwin, I., (1979). Chronic Parkinsonism in humans due to a product of meperidine-analog synthesis. *Science* **219** (1983), 979–980. http://www.ncbi.nlm.nih.gov/entrez/query.fcgi?cmd=Retrieve&db=PubMed&dopt=Citation&list_uids=6823561.
 58. Tetrud, J.W., Langston, J.W., (1989). MPTP-induced parkinsonism as a model for Parkinson's disease. *Acta Neurol. Scand. Suppl.* **126**, 35–40. http://www.ncbi.nlm.nih.gov/entrez/query.fcgi?cmd=Retrieve&db=PubMed&dopt=Citation&list_uids=2694734.
 59. Schapira, A.H., Cooper, J.M., Dexter, D., Jenner, P., Clark, J.B., Marsden, C.D., (1990). Mitochondrial complex I deficiency in Parkinson's disease. *J. Neurochem.* **1**, 1269. http://www.ncbi.nlm.nih.gov/entrez/query.fcgi?cmd=Retrieve&db=PubMed&dopt=Citation&list_uids=2154550.
 60. Valadas, J.S., Vos, M., Verstreken, P., (2015). Therapeutic strategies in Parkinson's disease: what we have learned from animal models. *Ann. N. Y. Acad. Sci.* **1338**, 16–37. <https://doi.org/10.1111/nyas.12577>.
 61. Przedborski, S., Jackson-Lewis, V., (1998). Mechanisms of MPTP toxicity. *Mov. Disord.* **13** (Suppl 1), 35–38. http://www.ncbi.nlm.nih.gov/entrez/query.fcgi?cmd=Retrieve&db=PubMed&dopt=Citation&list_uids=9613716.
 62. Manneschi, L., Dotti, M.T., Battisti, C., de Stefano, N., Federico, A., (1994). Muscle respiratory chain enzyme activities in parkinson's disease and in multisystem extrapyramidal disorders with parkinsonism as the main clinical feature. *Arch. Gerontol. Geriatr.* **19** (Suppl 1), 155–161. http://www.ncbi.nlm.nih.gov/entrez/query.fcgi?cmd=Retrieve&db=PubMed&dopt=Citation&list_uids=18649855.
 63. Jin, S.M., Lazarou, M., Wang, C., Kane, L.A., Narendra, D.P., Youle, R.J., (2010). Mitochondrial membrane potential regulates PINK1 import and proteolytic destabilization by PARL. *J. Cell Biol.* **191**, 933–942. http://www.ncbi.nlm.nih.gov/entrez/query.fcgi?cmd=Retrieve&db=PubMed&dopt=Citation&list_uids=21115803.
 64. Eldeeb, M.A., Esmaili, M., Hassan, M., Ragheb, M.A., (2022). The Role of PTEN-L in Modulating PINK1-Parkin-Mediated Mitophagy. *Neurotox. Res.* **40**, 1103–1114. <https://doi.org/10.1007/S12640-022-00475-W>.
 65. Morais, V.A., Haddad, D., Craessaerts, K., de Bock, P.-J., Swerts, J., Vilain, S., Aerts, L., Overbergh, L., et al., (2014). PINK1 Loss-of-Function Mutations Affect Mitochondrial Complex I Activity via NdufA10 Ubiquinone Uncoupling. *Science* **344**, 203–207. <https://doi.org/10.1126/science.1249161>.
 66. Vos, M., Esposito, G., Edirisinghe, J.N., Vilain, S., Haddad, D.M., Slabbaert, J.R., van Meensel, S., Schaap, O., et al., (1979). Vitamin K2 is a mitochondrial electron carrier that rescues pink1 deficiency. *Science* **336** (2012), 1306–1310. http://www.ncbi.nlm.nih.gov/entrez/query.fcgi?cmd=Retrieve&db=PubMed&dopt=Citation&list_uids=22582012.
 67. Vos, M., Lovisa, B., Geens, A., Morais, V.A., Wagnières, G., den van Bergh, H., Ginggen, A., de Strooper, B., et al., (2013). Near-infrared 808 nm light boosts complex IV-dependent respiration and rescues a Parkinson-related pink1 model. *PLoS One* **8**, e78562.
 68. Oueslati, A., Lovisa, B., Perrin, J., Wagnières, G., van den Bergh, H., Tardy, Y., Lashuel, H., (2015). Photobiomodulation Suppresses Alpha-Synuclein-Induced Toxicity in an AAV-Based Rat Genetic Model of Parkinson's Disease. *PLoS One* **10**, e0140880.
 69. Poole, A.C., Thomas, R.E., Andrews, L.A., McBride, H.M., Whitworth, A.J., Pallanck, L.J., (2008). The PINK1/Parkin pathway regulates mitochondrial morphology. *PNAS* **105**, 1638–1643. <https://doi.org/10.1073/pnas.0709336105>.
 70. Deng, H., Dodson, M.W., Huang, H., Guo, M., (2008). The Parkinson's disease genes pink1 and parkin promote mitochondrial fission and/or inhibit fusion in Drosophila. *PNAS* **105**, 14503–14508. <https://doi.org/10.1073/pnas.0803998105>.
 71. Han, H., Tan, J., Wang, R., Wan, H., He, Y., Yan, X., Guo, J., Gao, Q., et al., (2020). PINK1 phosphorylates Drp1S616 to regulate mitophagy-independent mitochondrial dynamics. *EMBO Rep.* **21**, e48686.
 72. Gao, Q., Tian, R., Han, H., Slone, J., Wang, C., Ke, X., Zhang, T., Li, X., et al., (2022). PINK1-mediated Drp1S616 phosphorylation modulates synaptic development and plasticity via promoting mitochondrial fission. *Signal Transd. Targeted Therapy* **7**, 1–16. <https://doi.org/10.1038/s41392-022-00933-z>.
 73. Vilain, S., Esposito, G., Haddad, D., Schaap, O., Dobrev, M.P., Vos, M., van Meensel, S., Morais, V.A., et al., (2012). The yeast complex I equivalent NADH dehydrogenase rescues pink1 mutants. *PLoS Genet.* **8**, e1002456.
 74. Narendra, D., Tanaka, A., Suen, D.F., Youle, R.J., (2009). Parkin-induced mitophagy in the pathogenesis of Parkinson disease. *Autophagy* **5**, 706–708. http://www.ncbi.nlm.nih.gov/entrez/query.fcgi?cmd=Retrieve&db=PubMed&dopt=Citation&list_uids=19377297.
 75. Narendra, D., Tanaka, A., Suen, D.-F., Youle, R.J., (2008). Parkin is recruited selectively to impaired mitochondria and promotes their autophagy. *J. Cell Biol.* **183**, 795–803. <https://doi.org/10.1083/jcb.200809125>.

76. Cornelissen, T., Vilain, S., Vints, K., Gounko, N., Verstreken, P., Vandenbergh, W., (2018). Deficiency of parkin and PINK1 impairs age-dependent mitophagy in *Drosophila*. *Elife* **7**, e35878.
77. Lee, J.J., Sanchez-Martinez, A., Zarate, A.M., Benincá, C., Mayor, U., Clague, M.J., Whitworth, A.J., (2018). Basal mitophagy is widespread in *Drosophila* but minimally affected by loss of Pink1 or parkin. *J. Cell Biol.* **217**, 1613–1622. <https://doi.org/10.1083/jcb.201801044>.
78. Beckham, T.H., Lu, P., Jones, E.E., Marrison, T., Lewis, C.S., Cheng, J.C., Ramshesh, V.K., Beeson, G., et al., (2013). LCL124, a Cationic Analog of Ceramide, Selectively Induces Pancreatic Cancer Cell Death by Accumulating in Mitochondria. *J. Pharmacol. Exp. Ther.* **344**, 167–178. <https://doi.org/10.1124/JPET.112.199216>.
79. Hou, Q., Jin, J., Zhou, H., Novgorodov, S.A., Bielawska, A., Szulc, Z.M., Hannun, Y.A., Obeid, L.M., et al., (2011). Mitochondrially targeted ceramides preferentially promote autophagy, retard cell growth, and induce apoptosis. *J. Lipid Res.* **52**, 278–288. <https://doi.org/10.1194/JLR.M012161>.
80. Novgorodov, S.A., Guduz, T.I., Obeid, L.M., (2008). Long-chain Ceramide Is a Potent Inhibitor of the Mitochondrial Permeability Transition Pore. *J. Biol. Chem.* **283**, 24707–24717. <https://doi.org/10.1074/JBC.M801810200>.
81. di Paola, M., Cocco, T., Lorusso, M., (2000). Ceramide interaction with the respiratory chain of heart mitochondria. *Biochemistry* **39**, 6660–6668. <https://doi.org/10.1021/Bi9924415>.
82. Raichur, S., Wang, S.T., Chan, P.W., Li, Y., Ching, J., Chaurasia, B., Dogra, S., Öhman, M.K., et al., (2014). CerS2 Haploinsufficiency Inhibits β -Oxidation and Confers Susceptibility to Diet-Induced Steatohepatitis and Insulin Resistance. *Cell Metab.* **20**, 687–695. <https://doi.org/10.1016/j.cmet.2014.09.015>.
83. Hammerschmidt, P., Ostkotte, D., Nolte, H., Gerl, M.J., Jais, A., Brunner, H.L., Sprenger, H.G., Awazawa, M., et al., (2019). CerS6-Derived Sphingolipids Interact with Mff and Promote Mitochondrial Fragmentation in Obesity. *Cell* **177**, 1536–1552.e23. <https://doi.org/10.1016/J.CELL.2019.05.008>.
84. Zigdon, H., Kogot-Levin, A., Park, J.-W., Goldschmidt, R., Kelly, S., Merrill, A.H., Scherz, A., Pewzner-Jung, Y., et al., (2013). Ablation of Ceramide Synthase 2 Causes Chronic Oxidative Stress Due to Disruption of the Mitochondrial Respiratory Chain. *J. Biol. Chem.* **288**, 4947–4956. <https://doi.org/10.1074/jbc.M112.402719>.
85. Vos, M., Klein, C., (2022). Ceramide-induced mitophagy impairs β -oxidation-linked energy production in PINK1 deficiency. *Autophagy* **18**, 703–704. <https://doi.org/10.1080/15548627.2022.2027193>.
86. Mandik, F., Vos, M., (2021). Neurodegenerative Disorders: Spotlight on Sphingolipids. *Int. J. Mol. Sci.* **22**. <https://doi.org/10.3390/IJMS222111998>.
87. Oleinik, N., Kim, J., Roth, B.M., Selvam, S.P., Gooz, M., Johnson, R.H., Lemasters, J.J., Ogretmen, B., (2019). Mitochondrial protein import is regulated by p17/PERMIT to mediate lipid metabolism and cellular stress. *Sci. Adv.* **5**. <https://doi.org/10.1126/SCIADV.AAX1978>.
88. Hernández-Corbacho, M.J., Salama, M.F., Canals, D., Senkal, C.E., Obeid, L.M., (1862). Sphingolipids in mitochondria, *Biochim Biophys Acta Mol Cell Biol. Lipids* **2017**, 56–68. <https://doi.org/10.1016/J.BBALIP.2016.09.019>.
89. Frank, S., Gaume, B., Bergmann-Leitner, E.S., Leitner, W.W., Robert, E.G., Catez, F., Smith, C.L., Youle, R.J., (2001). The role of dynamin-related protein 1, a mediator of mitochondrial fission, in apoptosis. *Dev. Cell* **1**, 515–525. [https://doi.org/10.1016/S1534-5807\(01\)00055-7](https://doi.org/10.1016/S1534-5807(01)00055-7).
90. Fugio, L.B., Coeli-Lacchini, F.B., Leopoldino, A.M., (2020). Sphingolipids and Mitochondrial Dynamic. *Cells*. **9** <https://doi.org/10.3390/CELLS9030581>.
91. Feng, Y., He, D., Yao, Z., Klionsky, D.J., (2014). The machinery of macroautophagy. *Cell Res.* **24**, 24–41. <https://doi.org/10.1038/CR.2013.168>.
92. Young, M.M., Wang, H.G., (2018). Sphingolipids as Regulators of Autophagy and Endocytic Trafficking. *Adv. Cancer Res.* **140**, 27–60. <https://doi.org/10.1016/BS.ACR.2018.04.008>.
93. Harvald, E.B., Olsen, A.S.B., Færgeman, N.J., (2015). Autophagy in the light of sphingolipid metabolism. *Apoptosis* **20**, 658–670. <https://doi.org/10.1007/S10495-015-1108-2>.
94. Egan, D.F., Shackelford, D.B., Mihaylova, M.M., Gelino, S., Kohnz, R.A., Mair, W., Vasquez, D.S., Joshi, A., et al., (2011). Phosphorylation of ULK1 (hATG1) by AMP-activated protein kinase connects energy sensing to mitophagy. *Science* **331**, 456–461. <https://doi.org/10.1126/SCIENCE.1196371>.
95. Lee, J.W., Park, S., Takahashi, Y., Wang, H.G., (2010). The association of AMPK with ULK1 regulates autophagy. *PLoS One* **5**. <https://doi.org/10.1371/JOURNAL.PONE.0015394>.
96. Russell, R.C., Tian, Y., Yuan, H., Park, H.W., Chang, Y. Y., Kim, J., Kim, H., Neufeld, T.P., et al., (2013). ULK1 induces autophagy by phosphorylating Beclin-1 and activating VPS34 lipid kinase. *Nature Cell Biol.* **15**, 741–750. <https://doi.org/10.1038/NCB2757>.
97. Obara, K., Ohsumi, Y., (2011). PtdIns 3-Kinase Orchestrates Autophagosome Formation in Yeast. *J. Lipids*. **2011**, 1–9. <https://doi.org/10.1155/2011/498768>.
98. Geng, J., Klionsky, D.J., (2008). The Atg8 and Atg12 ubiquitin-like conjugation systems in macroautophagy. “Protein modifications: beyond the usual suspects” review series. *EMBO Rep.* **9**, 859–864. <https://doi.org/10.1038/EMBOR.2008.163>.
99. Sou, Y.S., Waguri, S., Iwata, J.I., Ueno, T., Fujimura, T., Hara, T., Sawada, N., Yamada, A., et al., (2008). The Atg8 conjugation system is indispensable for proper development of autophagic isolation membranes in mice. *Mol. Biol. Cell* **19**, 4762–4775. <https://doi.org/10.1091/MBE.E08-03-0309/ASSET/IMAGES/LARGE/ZMK0110887640008.JPEG>.
100. Itakura, E., Kishi-Itakura, C., Mizushima, N., (2012). The hairpin-type tail-anchored SNARE syntaxin 17 targets to autophagosomes for fusion with endosomes/lysosomes. *Cell* **151**, 1256–1269. <https://doi.org/10.1016/J.CELL.2012.11.001>.
101. Kimura, S., Noda, T., Yoshimori, T., (2008). Dynein-dependent movement of autophagosomes mediates efficient encounters with lysosomes. *Cell Struct. Funct.* **33**, 109–122. <https://doi.org/10.1247/CSF.08005>.
102. Ravikumar, B., Acevedo-Arozena, A., Imarisio, S., Berger, Z., Vacher, C., O’Kane, C.J., Brown, S.D.M., Rubinsztein, D.C., (2005). Dynein mutations impair autophagic clearance of aggregate-prone proteins. *Nature Genet.* **37**, 771–776. <https://doi.org/10.1038/NG1591>.

103. Jäger, S., Bucci, C., Tanida, I., Ueno, T., Kominami, E., Saftig, P., Eskelinen, E.L., (2004). Role for Rab7 in maturation of late autophagic vacuoles. *J. Cell Sci.* **117**, 4837–4848. <https://doi.org/10.1242/JCS.01370>.
104. Nakamura, S., Yoshimori, T., (2017). New insights into autophagosome-lysosome fusion. *J. Cell Sci.* **130**, 1209–1216. <https://doi.org/10.1242/JCS.196352/265307/AM/NEW-INSIGHTS-INTO-AUTOPHAGOSOME-LYSOSOME-FUSION>.
105. Köchl, R., Hu, X.W., Chan, E.Y.W., Tooze, S.A., (2006). Microtubules facilitate autophagosome formation and fusion of autophagosomes with endosomes. *Traffic* **7**, 129–145. <https://doi.org/10.1111/J.1600-0854.2005.00368.X>.
106. Wijdeven, R.H., Janssen, H., Nahidiazar, L., Janssen, L., Jalink, K., Berlin, I., Neefjes, J., (2016). Cholesterol and ORP1L-mediated ER contact sites control autophagosome transport and fusion with the endocytic pathway. *Nature Commun.* **7** <https://doi.org/10.1038/NCOMMS11808>.
107. Narendra, D.P., Youle, R.J., (2011). Targeting mitochondrial dysfunction: role for PINK1 and Parkin in mitochondrial quality control. *Antioxid. Redox Signal.* **14**, 1929–1938 http://www.ncbi.nlm.nih.gov/entrez/query.fcgi?cmd=Retrieve&db=PubMed&dopt=Citation&list_uids=21194381.
108. Ahmed, I., Liang, Y., Schools, S., Dawson, V.L., Dawson, T.M., Savitt, J.M., (2012). Development and characterization of a new Parkinson's disease model resulting from impaired autophagy. *J. Neurosci.* **32**, 16503–16509. <https://doi.org/10.1523/JNEUROSCI.0209-12.2012>.
109. Hou, X., Watzlawik, J.O., Fiesel, F.C., Springer, W., (2020). Autophagy in Parkinson's Disease. *J. Mol. Biol.* **432**, 2651–2672. <https://doi.org/10.1016/J.JMB.2020.01.037>.
110. Dehay, B., Bové, J., Rodríguez-Muela, N., Perier, C., Recasens, A., Boya, P., Vila, M., (2010). Pathogenic lysosomal depletion in Parkinson's disease. *J. Neurosci.* **30**, 12535–12544. <https://doi.org/10.1523/JNEUROSCI.1920-10.2010>.
111. Alvarez-Erviti, L., Rodríguez-Oroz, M.C., Cooper, J.M., Caballero, C., Ferrer, I., Obeso, J.A., Schapira, A.H.V., (2010). Chaperone-mediated autophagy markers in Parkinson disease brains. *Arch. Neurol.* **67**, 1464–1472. <https://doi.org/10.1001/ARCHNEUROL.2010.198>.
112. Tanji, K., Mori, F., Kakita, A., Takahashi, H., Wakabayashi, K., (2011). Alteration of autophagosomal proteins (LC3, GABARAP and GATE-16) in Lewy body disease. *Neurobiol. Dis.* **43**, 690–697. <https://doi.org/10.1016/J.NBD.2011.05.022>.
113. Miura, E., Hasegawa, T., Konno, M., Suzuki, M., Sugeno, N., Fujikake, N., Geisler, S., Tabuchi, M., et al., (2014). VPS35 dysfunction impairs lysosomal degradation of α -synuclein and exacerbates neurotoxicity in a Drosophila model of Parkinson's disease. *Neurobiol. Dis.* **71**, 1–13.
114. Cullen, V., Lindfors, M., Ng, J., Paetau, A., Swinton, E., Kolodziej, P., Boston, H., Saftig, P., et al., (2009). Cathepsin D expression level affects alpha-synuclein processing, aggregation, and toxicity in vivo. *Mol. Brain* **2** <https://doi.org/10.1186/1756-6606-2-5>.
115. Spencer, B., Potkar, R., Trejo, M., Rockenstein, E., Patrick, C., Gindi, R., Adame, A., Wyss-Coray, T., et al., (2009). Beclin 1 gene transfer activates autophagy and ameliorates the neurodegenerative pathology in alpha-synuclein models of Parkinson's and Lewy body diseases. *J. Neurosci.* **29**, 13578–13588. <https://doi.org/10.1523/JNEUROSCI.4390-09.2009>.
116. Manzoni, C., Mamais, A., Dihanich, S., McGoldrick, P., Devine, M.J., Zerle, J., Kara, E., Taanman, J.W., et al., (2013). Pathogenic Parkinson's disease mutations across the functional domains of LRRK2 alter the autophagic/lysosomal response to starvation. *Biochem. Biophys. Res. Commun.* **441**, 862–866. <https://doi.org/10.1016/J.BBRC.2013.10.159>.
117. Higashi, S., Moore, D.J., Yamamoto, R., Minegishi, M., Sato, K., Togo, T., Katsuse, O., Uchikado, H., et al., (2009). Abnormal localization of leucine-rich repeat kinase 2 to the endosomal-lysosomal compartment in lewy body disease. *J. Neuropathol. Exp. Neurol.* **68**, 994–1005. <https://doi.org/10.1097/NEN.0B013E3181B44ED8>.
118. Matta, S., van Kolen, K., da Cunha, R., van den Bogaart, G., Mandemakers, W., Miskiewicz, K., de Bock, P.-J., Morais, V.A., et al., (2012). LRRK2 Controls an EndoA Phosphorylation Cycle in Synaptic Endocytosis. *Neuron* **75**, 1008–1021 <http://linkinghub.elsevier.com/retrieve/pii/S0896627312007593>.
119. Soukup, S., Kuenen, S., Vanhauwaert, R., Manetsberger, J., Hernández-Díaz, S., Swerts, J., Schoovaerts, N., Vilain, S., et al., (2016). A LRRK2-Dependent EndophilinA Phosphoswitch Is Critical for Macroautophagy at Presynaptic Terminals. *Neuron* **92**, 829–844.
120. Parekh, P., Sharma, N., Gadepalli, A., Shahane, A., Sharma, M., Khairnar, A., (2019). A Cleaning Crew: The Pursuit of Autophagy in Parkinson's Disease. *ACS Chem. Neurosci.* **10**, 3914–3926. <https://doi.org/10.1021/ACSCHEMNEURO.9B00244>.
121. Taniguchi, M., Kitatani, K., Kondo, T., Hashimoto-Nishimura, M., Asano, S., Hayashi, A., Mitsutake, S., Igarashi, Y., et al., (2012). Regulation of autophagy and its associated cell death by “sphingolipid rheostat”: reciprocal role of ceramide and sphingosine 1-phosphate in the mammalian target of rapamycin pathway. *J. Biol. Chem.* **287**, 39898–39910. <https://doi.org/10.1074/JBC.M112.416552>.
122. Lavie, G., Scarlatti, F., Sala, G., Carpentier, S., Levade, T., Ghidoni, R., Botti, J., Codogno, P., (2006). Regulation of autophagy by sphingosine kinase 1 and its role in cell survival during nutrient starvation. *J. Biol. Chem.* **281**, 8518–8527. <https://doi.org/10.1074/JBC.M506182200>.
123. Taniguchi, M., Okazaki, T., (2014). The role of sphingomyelin and sphingomyelin synthases in cell death, proliferation and migration—from cell and animal models to human disorders. *Biochim. Biophys. Acta (BBA) – Mol. Cell Biol. Lipids* **1841**, 692–703. <https://doi.org/10.1016/j.bbalip.2013.12.003>.
124. Guenther, G.G., Peralta, E.R., Rosales, K.R., Wong, S.Y., Siskind, L.J., Edinger, A.L., (2008). Ceramide starves cells to death by downregulating nutrient transporter proteins. *PNAS* **105**, 17402–17407. <https://doi.org/10.1073/PNAS.0802781105>.
125. Matarrese, P., Garofalo, T., Manganelli, V., Gambardella, L., Marconi, M., Grasso, M., Tinari, A., Misasi, R., et al., (2014). Evidence for the involvement of GD3 ganglioside in autophagosome formation and maturation. *Autophagy* **10**, 750–765. <https://doi.org/10.4161/AUTO.27959>.
126. Garofalo, T., Matarrese, P., Manganelli, V., Marconi, M., Tinari, A., Gambardella, L., Faggioni, A., Misasi, R., et al.,

- (2016). Evidence for the involvement of lipid rafts localized at the ER-mitochondria associated membranes in autophagosome formation. *Autophagy* **12**, 917–935. <https://doi.org/10.1080/15548627.2016.1160971>.
127. Toops, K.A., Tan, L.X., Jiang, Z., Radu, R.A., Lakkaraju, A., (2015). Cholesterol-mediated activation of acid sphingomyelinase disrupts autophagy in the retinal pigment epithelium. *Mol. Biol. Cell* **26**, 1–14. <https://doi.org/10.1091/MBE.E14-05-1028>.
 128. Giovedi, S., Ravanelli, M., Parisi, B., Bettegazzi, B., Guarnieri, F., (2020). Dysfunctional Autophagy and Endolysosomal System in Neurodegenerative Diseases: Relevance and Therapeutic Options. *Front. Cell. Neurosci.* **14**, 602116
 129. Andres-Alonso, M., Kreutz, M., Karpova, A., (2020). Autophagy and the endolysosomal system in presynaptic function. *Cell. Mol. Life Sci.*
 130. Kuijpers, M., Azarnia Tehran, D., Haucke, V., Soykan, T., (2020). The axonal endolysosomal and autophagic systems. *J. Neurochem.*
 131. Cullen, P.J., Steinberg, F., (2018). To degrade or not to degrade: mechanisms and significance of endocytic recycling. *Nature Rev. Mol. Cell Biol.* **19**, 679–696. <https://doi.org/10.1038/S41580-018-0053-7>.
 132. Naslavsky, N., Caplan, S., (2018). The enigmatic endosome - sorting the ins and outs of endocytic trafficking. *J. Cell Sci.* **131** <https://doi.org/10.1242/JCS.216499>.
 133. Burd, C., Cullen, P., (2014). Retromer: a master conductor of endosome sorting. *Cold Spring Harb Perspect Biol.* **6**, a016774
 134. Gallon, M., Cullen, P., (2015). Retromer and sorting nexins in endosomal sorting. *Biochem. Soc. Trans.* **43**, 33–47.
 135. Williams, E.T., Chen, X., Moore, D.J., (2017). VPS35, the Retromer Complex and Parkinson's Disease. *J. Parkinsons Dis.* **7**, 219–233. <https://doi.org/10.3233/JPD-161020>.
 136. Rojas, R., van Vlijmen, T., Mardones, G.A., Prabhu, Y., Rojas, A.L., Mohammed, S., Heck, A.J.R., Raposo, G., et al., (2008). Regulation of retromer recruitment to endosomes by sequential action of Rab5 and Rab7. *J. Cell Biol.* **183**, 513–526. <https://doi.org/10.1083/jcb.200804048>.
 137. Stenmark, H., Olkkonen, V.M., (2001). The Rab GTPase family REVIEWS3007–REVIEWS3007 *Genome Biol.* **2** <https://doi.org/10.1186/gb-2001-2-5-reviews3007>.
 138. Chen, K.E., Healy, M.D., Collins, B.M., (2019). Towards a molecular understanding of endosomal trafficking by Retromer and Retriever. *Traffic* **20**, 465–478. <https://doi.org/10.1111/TRA.12649>.
 139. Wang, J., Fedoseienko, A., Chen, B., Burstein, E., Jia, D., Billadeau, D.D., (2018). Endosomal receptor trafficking: Retromer and beyond. *Traffic* **19**, 578–590. <https://doi.org/10.1111/TRA.12574>.
 140. McMillan, K.J., Korswagen, H.C., Cullen, P.J., (2017). The emerging role of retromer in neuroprotection. *Curr. Opin. Cell Biol.* **47**, 72–82. <https://doi.org/10.1016/J.CEB.2017.02.004>.
 141. Lucas, M., Hierro, A., (2017). Retromer. *Curr. Biol.* **27**, R687–R689. <https://doi.org/10.1016/J.CUB.2017.05.072>.
 142. Kümmel, D., Ungermann, C., (2014). Principles of membrane tethering and fusion in endosome and lysosome biogenesis. *Curr. Opin. Cell Biol.* **29**, 61–66. <https://doi.org/10.1016/J.CEB.2014.04.007>.
 143. Luzio, J.P., Gray, S.R., Bright, N.A., (2010). Endosome-lysosome fusion. *Biochem. Soc. Trans.* **38**, 1413–1416. <https://doi.org/10.1042/BST0381413>.
 144. Luzio, J.P., Bright, N.A., Pryor, P.R., (2007). The role of calcium and other ions in sorting and delivery in the late endocytic pathway. *Biochem. Soc. Trans.* **35**, 1088–1091. <https://doi.org/10.1042/BST0351088>.
 145. Lauwers, E., Jacob, C., Andre, B., (2009). K63-linked ubiquitin chains as a specific signal for protein sorting into the multivesicular body pathway. *J. Cell Biol.* **185**, 493–502. <https://doi.org/10.1083/JCB.200810114>.
 146. Erpapazoglou, Z., Walker, O., Hagenauer-Tsapis, R., (2014). Versatile Roles of K63-Linked Ubiquitin Chains in Trafficking. *Cells* **3**, 1027–1088. <https://doi.org/10.3390/CELLS3041027>.
 147. Pfeffer, S., (2001). Membrane transport: retromer to the rescue. *Curr. Biol.* **11**, R109–R111.
 148. Seaman, M., (2005). Recycle your receptors with retromer. *Trends Cell Biol.* **15**, 68–75.
 149. Seaman, M.N.J., (2004). Cargo-selective endosomal sorting for retrieval to the Golgi requires retromer. *J. Cell Biol.* **165**, 111–122. <https://doi.org/10.1083/JCB.200312034>.
 150. Balderhaar, H.J.K., Arlt, H., Ostrowicz, C., Bröcker, C., Sündermann, F., Brandt, R., Babst, M., Ungermann, C., (2010). The Rab GTPase Ypt7 is linked to retromer-mediated receptor recycling and fusion at the yeast late endosome. *J. Cell Sci.* **123**, 4085–4094. <https://doi.org/10.1242/JCS.071977>.
 151. Harrison, M.S., Hung, C.S., Liu, T.T., Christiano, R., Walther, T.C., Burd, C.G., (2014). A mechanism for retromer endosomal coat complex assembly with cargo. *PNAS* **111**, 267–272. <https://doi.org/10.1073/PNAS.1316482111>.
 152. Priya, A., Kalaidzidis, I.v., Kalaidzidis, Y., Lambright, D., Datta, S., (2015). Molecular insights into Rab7-mediated endosomal recruitment of core retromer: deciphering the role of Vps26 and Vps35. *Traffic* **16**, 68–84. <https://doi.org/10.1111/TRA.12237>.
 153. Seaman, M.N.J., Harbour, M.E., Tattersall, D., Read, E., Bright, N., (2009). Membrane recruitment of the cargo-selective retromer subcomplex is catalysed by the small GTPase Rab7 and inhibited by the Rab-GAP TBC1D5. *J. Cell Sci.* **122**, 2371–2382. <https://doi.org/10.1242/JCS.048686>.
 154. Jimenez-Orgaz, A., Kvainickas, A., Nägele, H., Denner, J., Eimer, S., Dengjel, J., Steinberg, F., (2018). Control of RAB7 activity and localization through the retromer-TBC1D5 complex enables RAB7-dependent mitophagy. *EMBO J.* **37**, 235–254. <https://doi.org/10.15252/EMBJ.201797128>.
 155. Seaman, M.N.J., Mukadam, A.S., Breusegem, S.Y., (2018). Inhibition of TBC1D5 activates Rab7a and can enhance the function of the retromer cargo-selective complex. *J. Cell Sci.* **131** <https://doi.org/10.1242/JCS.217398>.
 156. Jia, D., Zhang, J.S., Li, F., Wang, J., Deng, Z., White, M. A., Osborne, D.G., Phillips-Krawczak, C., et al., (2016). Structural and mechanistic insights into regulation of the retromer coat by TBC1d5. *Nature Commun.* **7** <https://doi.org/10.1038/NCOMMS13305>.

157. Outeiro, T., Lindquist, S., (1979). Yeast cells provide insight into alpha-synuclein biology and pathobiology. *Science* **302** (2003), 1772–1775.
158. Cooper, A.A., Gitler, A.D., Cashikar, A., Haynes, C.M., Hill, K.J., Bhullar, B., Liu, K., Xu, K., et al., (2006). Alpha-synuclein blocks ER-Golgi traffic and Rab1 rescues neuron loss in Parkinson's models. *Science* **313**, 324–328. <https://doi.org/10.1126/science.1129462>.
159. Winslow, A.R., Chen, C.W., Corrochano, S., Acevedo-Arozena, A., Gordon, D.E., Peden, A.A., Lichtenberg, M., Menzies, F.M., et al., (2010). α -Synuclein impairs macroautophagy: implications for Parkinson's disease. *J. Cell Biol.* **190**, 1023–1037. <https://doi.org/10.1083/JCB.201003122>.
160. Yin, G., Lopes da Fonseca, T., Eisbach, S., Anduaga, A., Breda, C., Orcellet, M., Szegő, É., Guerreiro, P., et al., (2014). α -Synuclein interacts with the switch region of Rab8a in a Ser129 phosphorylation-dependent manner. *Neurobiol. Dis.* **70**, 149–161.
161. Dinter, E., Saridakis, T., Nippold, M., Plum, S., Diederichs, L., Komnig, D., Fensky, L., May, C., et al., (2016). Rab7 induces clearance of α -synuclein aggregates. *J. Neurochem.* **138**, 758–774.
162. Dodson, M.W., Zhang, T., Jiang, C., Chen, S., Guo, M., (2012). Roles of the Drosophila LRRK2 homolog in Rab7-dependent lysosomal positioning. *Hum. Mol. Genet.* **21**, 1350–1363. <https://doi.org/10.1093/hmg/ddr573>.
163. Jeong, G.R., Jang, E.H., Bae, J.R., Jun, S., Kang, H.C., Park, C.H., Shin, J.H., Yamamoto, Y., et al., (2018). Dysregulated phosphorylation of Rab GTPases by LRRK2 induces neurodegeneration. *Mol. Neurodegener.* **13**, 1–17. <https://doi.org/10.1186/S13024-018-0240-1/FIGURES/7>.
164. Lanning, N.J., Vanopstall, C., Goodall, M.L., Mackeigan, J.P., Looyenga, B.D., (2018). LRRK2 deficiency impairs trans-Golgi to lysosome trafficking and endocytic cargo degradation in human renal proximal tubule epithelial cells. *Am. J. Physiol. Renal Physiol.* **315**, F1465–F1477. <https://doi.org/10.1152/AJPRENAL.00009.2018/ASSET/IMAGES/LARGE/ZH20091886100006.JPEG>.
165. MacLeod, D.A., Rhinn, H., Kuwahara, T., Zolin, A., di Paolo, G., MacCabe, B.D., Marder, K.S., Honig, L.S., et al., (2013). RAB7L1 interacts with LRRK2 to modify intraneuronal protein sorting and Parkinson's disease risk. *Neuron* **77**, 425–439. <https://doi.org/10.1016/J.NEURON.2012.11.033>.
166. Eguchi, T., Kuwahara, T., Sakurai, M., Komori, T., Fujimoto, T., Ito, G., Yoshimura, S.I., Harada, A., et al., (2018). LRRK2 and its substrate Rab GTPases are sequentially targeted onto stressed lysosomes and maintain their homeostasis. *PNAS* **115**, E9115–E9124. <https://doi.org/10.1073/PNAS.1812196115>.
167. Liu, Z., Bryant, N., Kumaran, R., Beilina, A., Abeliovich, A., Cookson, M.R., West, A.B., (2018). LRRK2 phosphorylates membrane-bound Rabs and is activated by GTP-bound Rab7L1 to promote recruitment to the trans-Golgi network. *Hum. Mol. Genet.* **27**, 385–395. <https://doi.org/10.1093/HMG/DDX410>.
168. Purllyte, E., Dhekne, H.S., Sarhan, A.R., Gomez, R., Lis, P., Wightman, M., Martinez, T.N., Tonelli, F., et al., (2018). Rab29 activation of the Parkinson's disease-associated LRRK2 kinase. *EMBO J.* **37**, 1–18 <https://doi.org/10.15252/embj.201798099>.
169. Madero-Pérez, J., Fernández, B., Lara Ordóñez, A.J., Fdez, E., Lobbastael, E., Baekelandt, V., Hilfiker, S., (2018). RAB7L1-Mediated Relocalization of LRRK2 to the Golgi Complex Causes Centrosomal Deficits via RAB8A. *Front. Mol. Neurosci.* **11** <https://doi.org/10.3389/FNMOL.2018.00417>.
170. Gomez, R.C., Wawro, P., Lis, P., Alessi, D.R., Pfeffer, S. R., (2019). Membrane association but not identity is required for LRRK2 activation and phosphorylation of Rab GTPases. *J. Cell Biol.* **218**, 4157–4170. <https://doi.org/10.1083/JCB.201902184>.
171. Mir, R., Tonelli, F., Lis, P., Macartney, T., Polinski, N.K., Martinez, T.N., Chou, M.Y., Howden, A.J.M., et al., (2018). The Parkinson's disease VPS35[D620N] mutation enhances LRRK2-mediated Rab protein phosphorylation in mouse and human. *Biochem. J.* **475**, 1861–1883. <https://doi.org/10.1042/BCJ20180248>.
172. Dodson, M.W., Leung, L.K., Lone, M., Lizzio, M.A., Guo, M., (2014). Novel ethyl methanesulfonate (EMS)-induced null alleles of the Drosophila homolog of LRRK2 reveal a crucial role in endolysosomal functions and autophagy in vivo. *Dis. Model. Mech.* **7**, 1351–1363. <https://doi.org/10.1242/dmm.017020>.
173. Korolchuk, V.I., Schütz, M.M., Gómez-Llorente, C., Rocha, J., Lansu, N.R., Collins, S.M., Wairkar, Y.P., Robinson, I.M., et al., (2007). Drosophila Vps35 function is necessary for normal endocytic trafficking and actin cytoskeleton organisation. *J. Cell Sci.* **120**, 4367–4376. <https://doi.org/10.1242/JCS.012336>.
174. Kudo, N., Kumagai, K., Tomishige, N., Yamaji, T., Wakatsuki, S., Nishijima, M., Hanada, K., Kato, R., (2008). Structural basis for specific lipid recognition by CERT responsible for nonvesicular trafficking of ceramide. *PNAS* **105**, 488–493. <https://doi.org/10.1073/PNAS.0709191105>.
175. Hanada, K., Kumagai, K., Yasuda, S., Miura, Y., Kawano, M., Fukasawa, M., Nishijima, M., (2003). Molecular machinery for non-vesicular trafficking of ceramide. *Nature* **426**, 803–809. <https://doi.org/10.1038/NATURE02188>.
176. D'Angelo, G., Polishchuk, E., di Tullio, G., Santoro, M., di Campli, A., Godi, A., West, G., Bielawski, J., et al., (2007). Glycosphingolipid synthesis requires FAPP2 transfer of glucosylceramide. *Nature* **449**, 62–67. <https://doi.org/10.1038/NATURE06097>.
177. D'Angelo, G., Uemura, T., Chuang, C.C., Polishchuk, E., Santoro, M., Ohvo-Rekilä, H., Sato, T., di Tullio, G., et al., (2013). Vesicular and non-vesicular transport feed distinct glycosylation pathways in the Golgi. *Nature* **501**, 116–120. <https://doi.org/10.1038/NATURE12423>.
178. Capasso, S., Sticco, L., Rizzo, R., Pirozzi, M., Russo, D., Dathan, N.A., Campelo, F., Galen, J., et al., (2017). Sphingolipid metabolic flow controls phosphoinositide turnover at the trans-Golgi network. *EMBO J.* **36**, 1736–1754 <https://doi.org/10.15252/EMBJ.201696048>.
179. de Matteis, M.A., Luini, A., (2008). Exiting the Golgi complex. *Nature Rev. Mol. Cell Biol.* **9**, 273–284. <https://doi.org/10.1038/NRM2378>.
180. Klemm, R.W., Ejsing, C.S., Surma, M.A., Kaiser, H.J., Gerl, M.J., Sampaio, J.L., de Robillard, Q., Ferguson, C., et al., (2009). Segregation of sphingolipids and sterols during formation of secretory vesicles at the trans-Golgi network. *J. Cell Biol.* **185**, 601–612. <https://doi.org/10.1083/JCB.200901145>.

181. Deng, Y., Rivera-Molina, F.E., Toomre, D.K., Burd, C.G., (2016). Sphingomyelin is sorted at the trans Golgi network into a distinct class of secretory vesicle. *PNAS* **113**, 6677–6682. <https://doi.org/10.1073/PNAS.1602875113>.
182. Lin, G., Wang, L., Marcogliese, P.C., Bellen, H.J., (2019). Sphingolipids in the Pathogenesis of Parkinson's Disease and Parkinsonism. *Trends Endocrinol Metab* **30**, 106–117. <https://doi.org/10.1016/J.TEM.2018.11.003>.
183. Hicks, A.A., Pramstaller, P.P., Johansson, Å., Vitart, V., Rudan, I., Ugocsai, P., Aulchenko, Y., Franklin, C.S., et al., (2009). Genetic Determinants of Circulating Sphingolipid Concentrations in European Populations. *PLoS Genet.* **5**, e1000672.
184. Czubowicz, K., Jęśko, H., Wencel, P., Lukiw, W.J., Strosznajder, R.P., (2019). The Role of Ceramide and Sphingosine-1-Phosphate in Alzheimer's Disease and Other Neurodegenerative Disorders. *Mol. Neurobiol.* **56**, 5436–5455. <https://doi.org/10.1007/s12035-018-1448-3>.